

基于属性散射中心卷积核调制的SAR目标识别深层网络(中文/[English](#))

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摘要: 卷积神经网络(CNN)的特征提取能力与其参数量有关,一般来说,参数量越多,CNN的特征提取能力越强。但要学好这些参数需要大量的训练数据,而在实际应用中,可用于模型训练的合成孔径雷达(SAR)图像往往是有限的。减少CNN的参数量可以降低对训练样本的需求,但同时也会降低CNN的特征表达能力,影响其目标识别性能。针对此问题,该文提出一种基于属性散射中心(ASC)卷积核调制的SAR目标识别深层网络。由于SAR图像具有电磁散射特性,为了提取更符合SAR目标特性的散射结构和边缘特征,所提网络使用预先设定的具有不同指向和长度的ASC核对少量CNN卷积核进行调制以生成更多卷积核,从而在降低网络参数数量的同时保证其特征提取能力。此外,该网络在浅层使用ASC调制卷积核来提取更符合SAR图像特性的散射结构和边缘特征,而在高层使用CNN卷积核来提取SAR图像的语义特征。由于同时使用ASC调制卷积核和CNN卷积核,该网络能够兼顾SAR目标的电磁散射特性和CNN的特征提取优势。使用实测SAR图像进行的实验证明了所提网络可以在降低对训练样本需求的同时保证优秀的SAR目标识别性能。

关键词: 合成孔径雷达(SAR); 目标识别; 卷积神经网络(CNN); 属性散射中心(ASC); 卷积核调制

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Deep Network for SAR Target Recognition Based on Attribute Scattering Center Convolutional Kernel Modulation ([in English](#))

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Abstract: The feature extraction capability of Convolutional Neural Networks (CNNs) is related to the number of their parameters. Generally, using a large number of parameters leads to improved feature extraction capability of CNNs. However, a considerable amount of training data is required to effectively learn these parameters. In practical applications, Synthetic Aperture Radar (SAR) images available for model training are often limited. Reducing the number of parameters in a CNN can decrease the demand for training samples, but the feature expression ability of the CNN is simultaneously diminished, which affects its target recognition performance. To solve this problem, this paper proposes a deep network for SAR target recognition based on Attribute Scattering Center (ASC) convolutional kernel modulation. Given the electromagnetic scattering

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characteristics of SAR images, the proposed network extracts scattering structures and edge features that are more consistent with the characteristics of SAR targets by modulating a small number of CNN convolutional kernels using predefined ASC kernels with different orientations and lengths. This approach generates additional convolutional kernels, which can reduce the network parameters while ensuring feature extraction capability. In addition, the designed network uses ASC-modulated convolutional kernels at shallow layers to extract scattering structures and edge features that are more consistent with the characteristics of SAR images while utilizing CNN convolutional kernels at deeper layers to extract semantic features of SAR images. The proposed network focuses on the electromagnetic scattering characteristics of SAR targets and shows the feature extraction advantages of CNNs due to the simultaneous use of ASC-modulated and CNN convolutional kernels. Experiments based on the studied SAR images demonstrate that the proposed network can ensure excellent SAR target recognition performance while reducing the demand for training samples.

Key words: Synthetic Aperture Radar (SAR); Target recognition; Convolutional Neural Network (CNN); Attribute Scattering Center (ASC); Convolutional kernel modulation

1 引言

合成孔径雷达(Synthetic Aperture Radar, SAR)是一种有源微波传感器,其通过将一个小尺寸天线沿直线进行运动来合成等效的大尺寸天线,从而实现二维高分辨率成像。SAR不仅能够实现全天时、大范围地表的远距离高分辨率成像,且其成像不受光线、云雾等气候条件的限制,并且具有一定的地物穿透能力。因此,SAR技术已经在国防安全、农业生产估算、地质勘测等军民领域中发挥着日益重要的作用。作为SAR图像解译的重要步骤之一,SAR目标识别旨在对输入的目标区域判断其所属类别,其在民用和军事等诸多领域均具有不可估量的应用价值,受到了世界各国的广泛关注与深入研究,并取得了丰富的研究成果。

随着大规模数据集的出现和计算机处理能力的增长,深层网络在目标识别领域得到了广泛的研究与应用。为了模仿人脑的认知机制,深层网络构建了一个多层次的模型结构来实现多层非线性变换,从而将原始数据映射到更适合识别的特征空间来提高目标识别的性能。卷积神经网络(Convolutional Neural Network, CNN)^[1-6]作为一种典型的用于图像分类的深层网络,通过一个统一的框架实现特征提取和分类,具有强大的特征学习能力。现有研究表明,通过卷积运算学习到的深层特征对不同类型的目标具有更强的分辨力^[1]。目前,CNN已经成为SAR目标识别领域的一种主流方法。Chen等人^[7]构建了一个仅包含卷积层的SAR目标识别网络A-ConvNet,该方法通过去除全连接层降低了网络参数量,在一定程度上可以缓解CNN因训练数据不足而导致的过拟合问题。喻玲娟等人^[8]提出一种基于全卷积神经网络和改进卷积自编码器的SAR目标识别方法,具有较强的抗噪性能。赵鹏飞等人^[9]利

用EfficientNet^[4]提取的SAR目标的空间特征,并进一步利用循环神经网络提取图像序列的时序特征,实现了SAR目标的鲁棒识别。Huang等人^[10]提出了一种用于SAR目标识别的轻型CNN,其包括全局支路和局部支路,以结合多层次的特征进行目标识别。Liu等人^[11]设计了用于SAR图像识别的复卷积网络,并将SAR图像的电磁散射特性与浅层卷积层的初始化相结合,提升了识别效果。Zhang等人^[12]将传统手工梯度直方图特征与深度网络特征融合,设计了适用于舰船SAR目标的识别网络。Qosja等人^[13]将ConvNeXt^[14]用于SAR目标识别任务,并验证了部分数据扩充方法对于SAR数据的适用性。张翼鹏等人^[15]构建了提取视觉特征和拓扑特征的双分支卷积神经网络,用于小样本舰船分类任务。

上述基于CNN的SAR目标识别方法能够根据训练数据自动提取适合分类的特征,避免了传统手工特征需要依赖人工经验的缺点。但是,CNN的特征提取能力与其参数量有关,一般来说,参数量越多,CNN的特征提取能力越强,但需要大量的训练数据对参数进行训练。由于实际采集高质量的SAR图像需要昂贵的设备和资源,因此获取高质量的SAR图像相比于光学图像更加困难。此外,SAR图像的相干成像机制会导致一些问题,如SAR图像中目标会发生几何畸变和结构缺失、SAR图像对观测参数敏感等,使得建立完备的SAR目标识别数据集更为困难。因此,在实际应用中,用于SAR目标识别的训练样本数量往往有限,使得网络在训练过程中易发生严重的过拟合,从而影响网络的特征提取能力,导致识别性能的下降。为了解决该问题,一方面,可以使用数据扩充技术,然而现有方法大多将光学领域的的数据扩充方法直接应用于SAR图像,这种做法的适用性难以保证,可能会产生不符合SAR图像特性的扩充结果,影响SAR目标识别性

能。另一方面,减少CNN的参数量可以降低对训练样本的需求,但同时也会降低CNN的特征表达能力,影响其目标识别性能。因此,设计一个适用于有限训练数据量、能够有效提取SAR图像特征的SAR目标识别深层网络具有重要意义。

为直观理解CNN从图像中学习的具体内容,文献[16]以在ImageNet数据集上训练的AlexNet[4]为例,对其第1层的卷积核进行可视化,直观展示了第1层卷积核所提取的线性结构和边缘特征。而SAR图像是基于电磁后向散射原理得到的,与光学图像相比,其具有离散、不连续性[17],其在图像中的显著结构包含散射结构和边缘特征,因此对SAR图像提取其中的散射结构和边缘特征时需要结合其电磁散射特性进行考虑。属性散射中心(Attributed Scattering Center, ASC)模型[18,19]采用7个参数描述每个散射中心的特性,蕴藏了丰富的物理以及几何属性,精确刻画了目标的真实散射机理,能够准确建模高分辨SAR图像的目标散射特性,因此,本文提出一种基于ASC卷积核调制的SAR目标识别深层网络,旨在通过结合SAR目标的电磁散射特性设计更简单的网络结构来提取更符合SAR图像特性的特征,从而在降低网络参数量的同时保证网络的特征提取能力。具体来说,所提方法在网络的浅层使用了ASC调制卷积核,用以提取更符合SAR图像特性的散射结构和边缘特征,而在网络的高层使用CNN本身的卷积核,用以提取SAR图像的语义特征。其中,ASC调制卷积核使用预先设定的具有不同指向和长度的ASC核对CNN本身的卷积核进行调制,该操作可以利用少量可学习卷积核生成更多的卷积核,从而在降低网络参数量的同时保证网络的特征提取能力。由于整个网络同时使用了ASC调制卷积核和CNN本身的卷积核,因此该网络能够兼顾SAR目标的电磁散射特性和CNN的特征提取优势,进一步降低对训练样本的需求。基于MSTAR实测数据集的实验结果表明,本文所设计的网络可以在降低对训练样本需求的同时保证优秀的目标识别性能。

2 属性散射中心模型

由于SAR图像是基于电磁后向散射原理得到的,与光学图像相比,其具有独特的电磁散射特性。当雷达工作在高频区时,可以近似认为分布式目标的散射场是由构成该目标的各散射中心的散射场累加而成的[18]。ASC模型是一种经典的散射中心模型,其描述了SAR目标的后向散射对频率和方位的依赖性,其模型参数包含丰富的物理和几何属

性,可以更准确地表征雷达目标在高频区的电磁散射特性,因而得到了广泛的关注与应用。分布式目标在高频区的散射场可以表示为

$$E(f, \varphi; \Theta) = \sum_{i=1}^K E_i(f, \varphi; \theta_i) \quad (1)$$

其中, f 代表雷达频率; φ 代表方位角; Θ 代表散射中心的参数集; K 代表目标中包含的散射中心总数。对于一个ASC,其散射场可以用式(2)的ASC模型来表示[18]:

$$\begin{aligned} E_i(f, \varphi; \theta_i) = & A_i \cdot \left(j \frac{f}{f_c} \right)^{\alpha_i} \\ & \cdot \exp \left(\frac{-j4\pi f}{c} (x_i \cos \varphi + y_i \sin \varphi) \right) \\ & \cdot \text{sinc} \left(\frac{2\pi f}{c} L_i \sin(\varphi - \bar{\varphi}_i) \right) \\ & \cdot \exp(-2\pi f \gamma_i \sin \varphi) \end{aligned} \quad (2)$$

其中, f_c 表示雷达的中心频率; j 表示虚数单位; c 代表电磁波的传播速度,即光速; $\text{sinc}(\cdot) = \sin(\cdot)/(\cdot)$ 表示sinc函数; 第 i 个ASC的属性参数可以表示为 $\theta_i = [A_i, \alpha_i, x_i, y_i, L_i, \bar{\varphi}_i, \gamma_i]$ ($i = 1, 2, \dots, K$); A_i 表示后向散射系数,为复数值; α_i 表示频率依赖因子,描述了ASC的后向散射幅度对频率的依赖性; x_i 和 y_i 分别表示ASC在距离维和方位维上的位置坐标; L_i 表示分布式ASC的长度; $\bar{\varphi}_i$ 表示分布式ASC的指向; γ_i 表示方位依赖因子,描述了ASC的后向散射幅度对方位的依赖性。若第 i 个ASC为局部式ASC,则其满足 $L_i = \bar{\varphi}_i = 0$; 若第 i 个ASC为分布式ASC,则其满足 $\gamma_i = 0, L_i > 0$ 。

3 方法介绍

3.1 网络结构与流程

图1展示了本文设计的基于ASC卷积核调制的SAR目标识别深层网络的结构图。从图1可以看出,该网络包含5个卷积层和2个全连接层。由于SAR图像是基于电磁后向散射原理得到的,与光学图像相比,具有独特的电磁散射特性,因此对SAR图像提取特征时应该考虑其电磁散射特性,以提取更符合SAR目标特性的特征。考虑到CNN的浅层负责提取图像中的低级结构特征,所提方法在前两个卷积层使用本文设计的ASC调制卷积核,以提取更符合SAR图像特性的散射结构和边缘特征。后3个卷积层依然使用CNN卷积核,以保证提取更具鉴别性的语义特征。在经过5个卷积层后,所提方法将提取的卷积特征经过两个全连接层映射为与类别数相同的维度,然后将其送入softmax层,完成

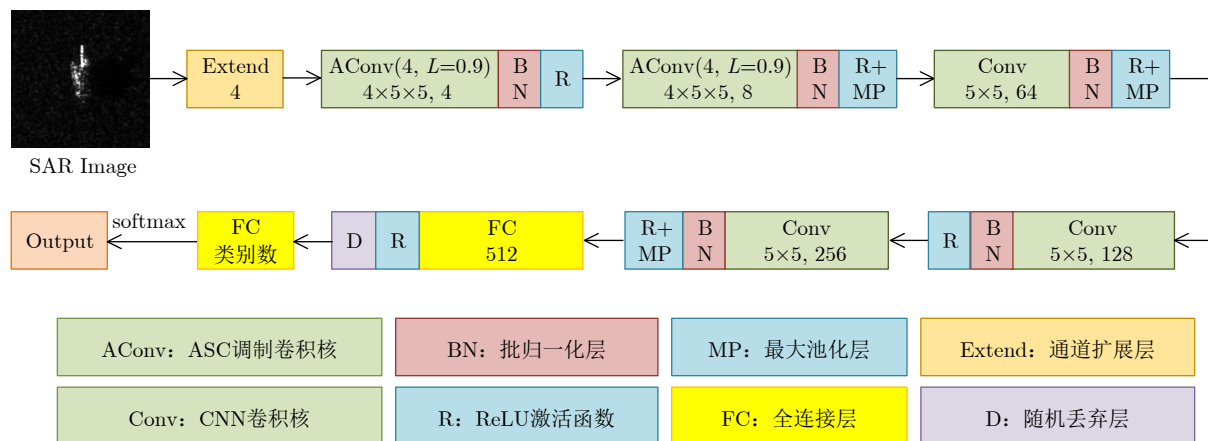


图1 基于ASC卷积核调制的SAR目标识别深层网络的结构图

Fig. 1 Architecture of the deep network for SAR target recognition based on ASC convolutional kernel modulation

对输入图像的识别。整个网络使用交叉熵损失函数以端到端的方式进行训练，并使用随机梯度下降法来更新网络参数。

接下来，本文将对所设计的基于ASC卷积核调制的SAR目标识别深层网络中的ASC调制卷积核的生成过程、ASC调制卷积核的卷积过程以及ASC调制卷积核的参数更新过程分别进行详细介绍。

3.2 ASC调制卷积核的生成过程

由于SAR图像是基于电磁后向散射原理得到的，与光学图像相比，其具有独特的电磁散射特性。当雷达工作在高频区时，可以近似认为分布式目标的散射场是由构成该目标的各散射中心的散射场累加而成的^[18]。因此，为了提取更符合SAR图像特性的散射结构和边缘特征，本文设计了ASC调制卷积核。该卷积核是用预先设定好的一组ASC核对CNN的可学习卷积核进行调制得到的，这样可以在只学习一小组CNN卷积核的情况下，通过ASC调制得到更多的卷积核，从而可以在降低参数量的同时保证CNN的特征提取能力。

为了得到ASC调制卷积核，首先需要预先生成一组ASC核。由第2节可知，每个ASC都包含7个属性参数，表示为 $\theta = [A, \alpha, x, y, L, \bar{\varphi}, \gamma]$ 。为了生成ASC核，本文首先将每个ASC置于核中心，即令 $x = y = 0$ ，然后将散射中心的幅度归一化，即令 $A = 1$ 。由于所提方法希望用ASC提取更适合SAR图像的散射结构和边缘特征，因此本文采用分布式ASC，即令 $\gamma = 0, L > 0$ 。另外，文献^[20]对ASC模型的参数进行了分析，对于高分辨率SAR图像，其带宽相对于载频通常较小，此时 α 对目标后向散射场的幅度影响很小，所以可以将 α 近似为0。此时，只剩下 L 和 $\bar{\varphi}$ 这两个参数待确定，其中， L 表示

分布式ASC的长度， $\bar{\varphi}$ 表示分布式ASC的指向。为了从SAR图像中提取不同方向的散射结构和边缘特征，本文令 $\bar{\varphi}$ 在 $[-90^\circ, 90^\circ]$ 之间均匀取值，假设共取 M 个值，则可以得到 M 个不同指向的分布式ASC。而 L 的取值则由图像的空间分辨率以及CNN的卷积核尺寸共同决定，其中空间分辨率表示输入图像中相邻两个像素点所代表的实际距离。假设空间分辨率为 m ，卷积核的尺寸为 $r \times r$ ，则 $L = p \times m$ ($p = 1, 2, \dots, r - 1$)。

在CNN中，卷积核尺寸越大，其感受野越大，对于SAR图像来说，大的感受野可以在一定程度上缓解斑点噪声的影响，但是卷积核尺寸越大也会增加网络待学习的参数量，因此在本文设计的网络中，所有卷积层都采用 5×5 的卷积核。以MSTAR数据集^[21]为例，其空间分辨率为 $0.3 \text{ m} \times 0.3 \text{ m}$ ，因此，对于 5×5 的卷积核， L 的取值可以为 $[0.3, 0.6, 0.9, 1.2]$ 。考虑到 $\bar{\varphi}$ 的取值越多，得到的分布式ASC的指向越丰富，但同时也会增加网络的参数量，因此本文综合考虑参数量和指向丰富性，令 $M = 4$ ，即 $\bar{\varphi} = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$ 。根据 L 和 $\bar{\varphi}$ 的组合，可以得到一组 5×5 的ASC核，如图2所示。由于所提方法希望通过ASC调制核提取更符合SAR图像特性的散射结构和边缘特征，因此预设的ASC核中的分布式ASC的长度不应太短，同时又为了保证ASC核中分布式ASC的完整性，结合图2的可视化结果，最终选择 $L = 0.9$ ， $\bar{\varphi} = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$ 对应的ASC核作为预设的一组ASC核，如图2中红框所示。

由于最终选定的ASC核是一组具有不同指向的ASC核，为了将指向信息编码进ASC调制核中，所提方法采用三维的CNN卷积核进行调制。三维CNN卷积核的尺寸可以表示为 $N \times W \times W$ ，其中

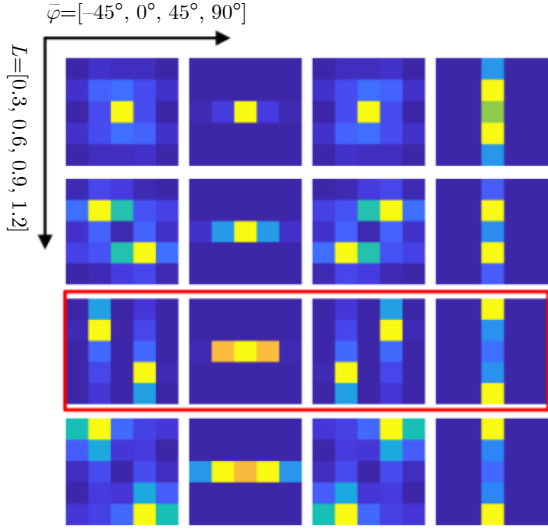


图2 根据预设的 L 和 $\bar{\varphi}$ 的组合得到的一组 5×5 的ASC核
(红框内的ASC核为最终选定的ASC核)

Fig. 2 A set of ASC kernels obtained according to the combinations of L and $\bar{\varphi}$ (the ASC kernels within the red box are the final selected ASC kernels)

$W \times W$ 表示卷积核的尺寸, N 表示其通道数。结合前面的分析, 本文采用的三维CNN卷积核的 $W = 5$, $N = 4$ 。一个ASC调制卷积核是用一组预设的具有不同指向的ASC核对可学习的三维CNN卷积核调制而成, 用图2中红框选中的ASC核对三维CNN卷积核的具体调制过程如图3和式(3)所示。

$$\mathbf{AC}_{i,m} = \mathbf{C}_i \circ \mathbf{A}(m) \quad (3)$$

其中, $\mathbf{AC}_{i,m}$ 表示第 i 个ASC调制卷积核的第 m 个方向通道, $m = 1, 2, 3, 4$; $\mathbf{C}_i$ 表示第 i 个可学习的三维CNN卷积核; $\mathbf{A}(m)$ 表示预设的一组具有不同指向的ASC核(对复数ASC核进行取模操作得到实数ASC核)中的第 m 个ASC核, 其表达式如式(2)所示; $\circ$ 表示 $\mathbf{A}(m)$ 与 $\mathbf{C}_i$ 中的每个通道对应元素相乘。第 i 个ASC调制卷积核又可以表示为

$$\mathbf{AC}_i = (\mathbf{AC}_{i,1}, \mathbf{AC}_{i,2}, \mathbf{AC}_{i,3}, \mathbf{AC}_{i,4}) \quad (4)$$

由式(4)可以看出, 第 i 个ASC调制卷积核 $\mathbf{AC}_i$ 实际上是4个三维卷积核, 如图3所示。对于一个ASC调制卷积核, 其尺寸可以表示为 $U \times N \times W \times W$, 其中 U 表示指向通道数, 本文得到的ASC调制卷积核的 $U = 4, N = 4, W = 5$ 。但是, 对于每一个ASC调制卷积核, 由于ASC核为预先设定好的, 因此只需要更新其中可学习的三维CNN卷积核的参数, 即 $N \times W \times W$, 这意味着通过这种调制过程, 所提方法可以在不增加参数量的前提下学习到更适合SAR目标的描述能力更强的特征。且预设的ASC核本身具有多个指向, 因此ASC调制卷

积核经过训练可以自动提取不同方位角下目标图像特征用于识别。

3.3 ASC调制卷积核的卷积过程

在本文设计的网络中, ASC调制卷积核也用来生成特征图, 与传统CNN卷积核生成的特征图相比, ASC调制卷积核能够提取更符合SAR图像特性的散射结构和边缘特征。由于ASC调制卷积核是4个三维卷积核, 因此对于一幅单通道的二维输入图像, 应先将进行通道扩展, 变成 $1 \times 4 \times S_i \times S_i$ 的形式, 其中1表示输入图像为单通道, 4表示扩展通道数, 扩展后的通道将用来对方向信息进行编码, 因此将其称为方向通道, S_i 表示输入图像的尺寸。对于一幅输入图像, 使用一个ASC调制卷积核生成的特征图 $\hat{\mathbf{F}}$ 可以表示为

$$\hat{\mathbf{F}} = \text{AConv}(\mathbf{F}, \mathbf{AC}_i) \quad (5)$$

其中, AConv表示ASC调制卷积操作, $\mathbf{F}$ 表示输入图像; $\mathbf{AC}_i$ 表示第 i 个ASC调制卷积核。式(5)表示的卷积过程如图4所示, S_o 表示输出图像的尺寸。输出特征图 $\hat{\mathbf{F}}$ 的第 k 个通道可以表示为

$$\hat{\mathbf{F}}_k = \sum_{n=1}^4 \mathbf{F}^{(n)} * \mathbf{AC}_{i,m}^{(n)} \quad (6)$$

其中, $\mathbf{F}^{(n)}$ 表示 $\mathbf{F}$ 的第 n 个方向通道; $\mathbf{AC}_{i,m}^{(n)}$ 表示第 i 个ASC调制卷积核的第 m 个方向通道中的第 n 个核; $*$ 表示卷积操作。由式(6)可以看出, 输出特征

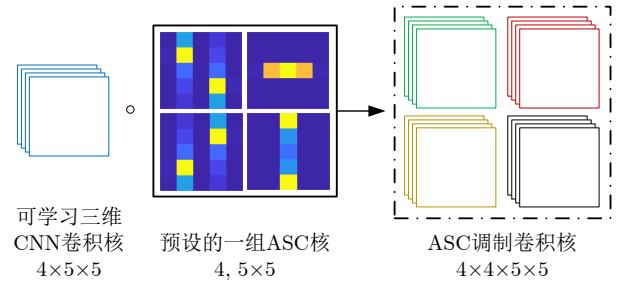


图3 ASC调制卷积核的生成过程

Fig. 3 The generation process of ASC modulated convolutional kernel

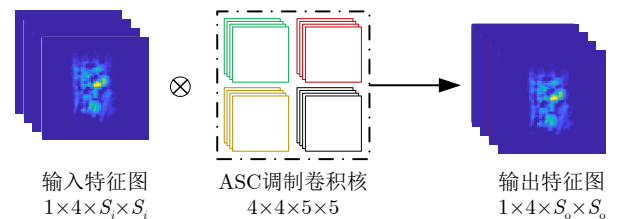


图4 ASC调制卷积核的卷积过程

Fig. 4 The convolution process of ASC modulated convolutional kernel

图 $\hat{F}$ 的每个通道都对应了预设的ASC核中不同的指向,通过这种卷积操作,所提方法可以将指向信息编码进特征图中。

由于在网络中,每个卷积层的输入一般均为多通道特征图,因此,为了表示一个使用ASC调制卷积核的卷积层的完整卷积过程,本节假设某个卷积层的输入特征图的维度为 $5 \times 4 \times S_i \times S_i$,其中5表示输入特征图为5通道,4表示方向通道数, S_i 表示输入特征图的尺寸;假设该卷积层共使用10组ASC调制卷积核,由于卷积核的通道数需与输入特征图的通道数相等,因此该卷积层的卷积核维度为 $10 \times 5 \times 4 \times 4 \times 5 \times 5$,其中10表示卷积核的组数,5表示卷积核的通道数, $4 \times 4 \times 5 \times 5$ 表示每个ASC调制卷积核的维度。由于预设ASC的调制作用,该层卷积层的可学习参数为 $10 \times 5 \times (4 \times 5 \times 5 + 1)$,仅为传统卷积层可学习参数量的四分之一。该卷积层对应的完整卷积过程如图5所示, S_o 表示输出特征图的尺寸。

3.4 ASC调制卷积核的参数更新过程

虽然ASC调制卷积核是用ASC核对CNN卷积核调制得到的,但由于ASC核是预先设定好的,因此在反向传播过程中,只需要更新其中可学习的CNN卷积核 C_i 的参数。与传统CNN卷积核的更新类似,ASC调制卷积核的更新准则也采用随机梯度下降和链式法则,其参数更新公式如下所示:

$$\delta = \frac{\partial L}{\partial C_i} = \sum_{m=1}^4 \frac{\partial L}{\partial C_{i,m}} \circ AC_{i,m} \quad (7)$$

$$C_i = C_i - \eta \delta \quad (8)$$

其中, L 表示网络的损失函数; η 表示网络的学习率。

4 实验结果与分析

本节首先介绍实验中采用的数据集、评价准则以及部分相关的实验细节;然后将所提方法与一些现有的基于深度学习的SAR目标识别方法的性能进行比较,以验证所提方法的有效性;最后进行了模型分析,包括预设的ASC核的相关参数对识别性能的影响以及不同训练样本数对识别性能的影响。所有实验代码均基于Ubuntu系统下的Pytorch进行编写。

4.1 实验数据集介绍

本文中SAR目标识别实验采用的数据集是MSTAR实测数据集^[21]和OpenSARShip舰船数据集^[22]。MSTAR数据集由工作在X波段的雷达采集得到,包含10类不同的地面军事目标,每幅图像都为复值SAR图像,其分辨率为 $0.3 \text{ m} \times 0.3 \text{ m}$,图像尺寸为128像素 $\times$ 128像素。图6展示了10类不同地面目标的SAR图像及其对应的光学图像示例。与光学图像相比,SAR图像的视觉可分性较差。

MSTAR数据集中SAR图像的方位角为 $0^\circ \sim 360^\circ$,训练集由俯仰角为 17° 的SAR图像构成,测试集由俯仰角为 15° 的SAR图像构成。参考文献^[23-25],多数SAR目标识别方法都是基于3类7型子数据集和10类14型子数据集进行验证,这两类子数据集的具体信息分别如表1和表2所示。在这两类子数据集

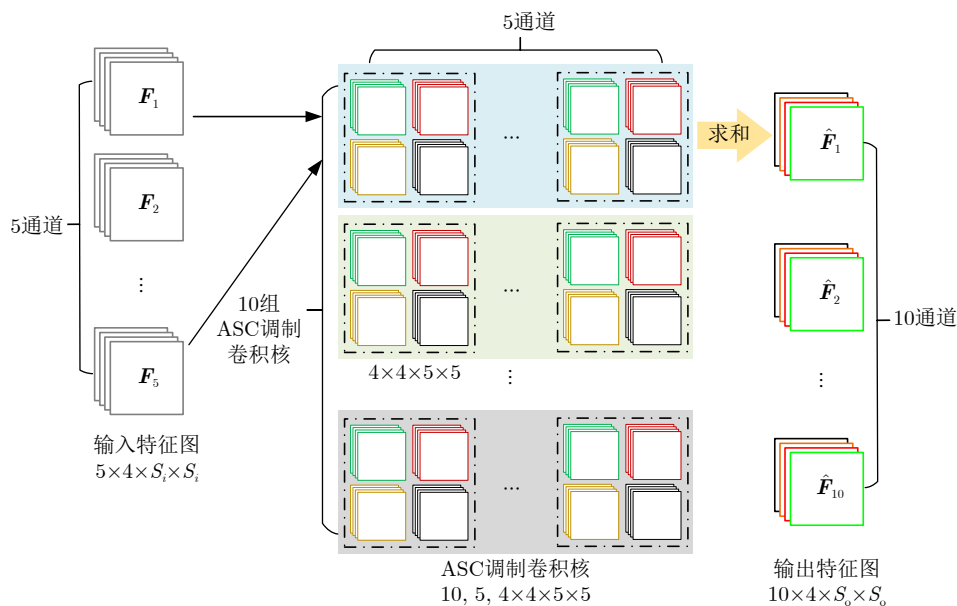


图5 使用ASC调制卷积核的卷积层中的完整卷积过程示例

Fig. 5 An example of the complete convolution process in a convolutional layer using ASC modulated convolution kernels



图 6 MSTAR 10类不同地面目标的SAR图像及其对应的光学图像示例

Fig. 6 Examples of SAR images and corresponding optical images of ten different types of ground targets in MSTAR dataset

中，测试样本包含部分未出现在训练样本中的变体型号，导致其识别难度更大，对于模型的特征提取能力要求更高。

OpenSARShip舰船数据集包含了11346张从哨兵1号SAR图像中截取的舰船切片，如图7所示，该数据集的背景更为复杂，包含海杂波、舰船尾迹、干扰船只、礁石等背景干扰等，且舰船目标本身具有类内差异大、类间差异相对小的特性，使得OpenSARShip数据集对于识别方法更具挑战性，更加考验模型的特征提取能力。

表 1 MSTAR的3类7型子数据集的具体信息

Tab. 1 The detailed information of three-target MSTAR data

类别	型号	训练样本数	测试样本数
BMP2	SNC21	233	196
	SN9563	0	195
	SN9566	0	196
BTR70	C71	233	196
	SNS7	0	191
T72	SN132	232	196
	SN812	0	195

OpenSARShip舰船数据集包含两类图像产品，本文选取了地距影像(Ground Range Detected, GRD)用于实验，其空间分辨率为 $10\text{ m} \times 10\text{ m}$ 。由于各类别样本数量极不均衡，因此本文选取了样本数量较多的Cargo, Fishing, Tanker类别用于识别实验，用于识别的3类数据集信息如表3所示。

表 2 MSTAR的10类14型子数据集的具体信息

Tab. 2 The detailed information of ten-target MSTAR data

类别	训练样本数	测试样本数
BTR60	255	195
2S1	299	274
BRDM2	298	274
D7	299	274
T62	299	273
ZIL131	299	274
ZSU23/4	299	274
BMP2	233	587
BTR70	233	196
T72	232	582

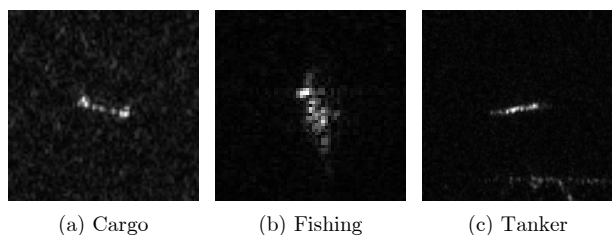


图7 OpenSARShip舰船目标SAR图像示例

Fig. 7 Examples of SAR images in OpenSARShip dataset

表3 OpenSARShip的3类数据集具体信息
Tab. 3 The detailed information of three-target
OpenSARShip data

类别	训练样本数	测试样本数
Cargo	241	159
Fishing	92	23
Tanker	115	76

4.2 评价准则和实验细节

为了验证本文设计的基于ASC卷积核调制的SAR目标识别深层网络的有效性,本文采用正确分类的概率(Probability of Correct Classification, PCC)来衡量各方法的目标识别性能,其计算公式如下:

$$PCC = \frac{\sum_{i=1}^{N_c} Tr_i}{\sum_{i=1}^{N_c} Q_i} \quad (9)$$

其中, Tr_i 表示第 i 类测试样本中被正确识别的样本数; Q_i 表示第 i 类测试样本的总数; N_c 表示类别总数。PCC值越大表示识别性能越好。

本文所设计的网络采用小批量随机梯度下降法进行训练,批尺寸为16,动量参数设为0.9,权值衰减因子设为0.0005,初始学习率设为0.005,且每隔100个epoch将学习率变为原来的0.1倍。训练epoch总数为500,以确保网络完全收敛。

4.3 识别性能比较

为了验证本文所设计的基于ASC卷积核调制的SAR目标识别深层网络的有效性,本节将其与一些现有的基于深度学习的SAR目标识别方法的性能进行比较,包括文献[7]提出的A-ConvNet、文献[2]提出的VGG16、文献[3]提出的ResNet34以及文献[26]提出的基于部件解析的多尺度SAR目标识别网络(CA-MCNN)。其中A-ConvNet, VGG16以及ResNet34是3种基于CNN的SAR目标识别方法,但在原始文献中,为了得到更好的性能,这两种方法

都采用了数据扩充技术。然而,现有的应用于SAR图像的数据扩充方法往往是直接采用了光学图像领域中的数据扩充方法,这种做法难以保证对SAR图像的适用性。同时,在实际应用中,选择何种扩充方式以及样本扩充数量难以确定。这些问题可能会导致不合适的扩充结果,从而影响SAR目标识别性能。相比之下,本文设计的网络并不存在这些问题,因为所提方法不需要通过数据扩充技术来保证识别性能。因此,为了保证公平性,本文根据文献[2,3,7]提供的网络结构和参数,在不进行数据扩充的条件下得到这3种方法的识别结果,并将其与本文所提方法的识别结果进行比较。CA-MCNN将SAR目标的ASC视为其部件,利用网络将由整幅SAR图像提供的全局信息与对应于目标ASC的部件信息相结合以得到更加稳健且描述能力更强的特征,同时,该网络还利用了多尺度特征图的融合以增强全局特征图的描述能力,从而进一步增强最终得到的融合特征的描述能力。此外,本节将与本文所设计的网络具有相同网络结构,但所有卷积层都使用传统的二维CNN卷积核的网络作为BaseNet,同样将其性能与本文所设计网络的性能进行对比。相比于图1中的所提方法,BaseNet不需要进行通道扩展,且前两层卷积层的卷积核个数等于对应层的ASC调制卷积核组数乘以方向通道数,以获得与所提方法相同数量的特征图。

本节分别基于3类7型MSTAR子数据集、10类14型MSTAR子数据集和OpenSARShip 3类舰船数据集对比了所提网络与上述方法的识别性能。

4.3.1 基于3类7型MSTAR子数据集的识别实验

本节将讨论所设计的基于ASC卷积核调制的SAR目标识别网络对3类7型MSTAR子数据集的有效性,表4展示了不同方法对3类7型MSTAR子数据集的识别性能以及各方法的参数量。

从表4可以看出,本文设计的网络的PCC可以达到0.9875。首先,所提方法的PCC高于VGG16和ResNet34,这是由于VGG16和ResNet34的参数量较大,需要利用大量的数据进行训练,但本文使用的3类7型子数据集中包含的训练样本数量有限,且实验并未采用数据扩充技术,因此无法保证VGG16和ResNet34的特征提取能力。另外,在3类7型子数据集中,其测试数据中包含未在训练数据中出现过的变体型号,这进一步增加了对该数据集识别的难度,因此,VGG16和ResNet34对3类7型子数据集的识别性能并不理想。而本文设计的网络具有更少的参数,其对训练样本的需求更低,因此

更适合训练样本数量有限的3类7型MSTAR子数据集。其次，所提方法的PCC也高于A-ConvNet的PCC，这是由于CNN的特征提取能力与其参数量有关，一般来说，参数量越多，CNN的特征提取能力越强，因此虽然A-ConvNet具有最少的参数，其对训练样本的需求相对更低，但过少的参数也降低了其特征提取能力，导致其对3类7型MSTAR子数据集的识别性能依然不够理想。而本文设计的网络使用了ASC调制卷积核，能够提取更符合SAR图像特性的散射结构和边缘特征，因此其能够取得更好的识别性能。此外，与BaseNet相比，本文设计的网络不仅具有更少的参数，同时也具有更高的PCC，这是因为所提方法使用的ASC调制卷积核可以在只学习一小组CNN卷积核的情况下，通过ASC调制得到更多的卷积核，从而可以在降低参数数量的同时保证CNN的特征提取能力，且ASC调制可以有效提升网络的识别效果，其作用不会随着迭代的过程消失。最后，从表4可以看出，由于能够利用SAR目标的图像信息和ASC信息来学习描述能力更强的特征，CA-MCNN的PCC高于其他对比方法，但与方法相比，本文设计的网络不仅不需要对每张SAR图像提取ASC，而且具有更少的参数和更高的PCC，这也说明本文设计的基于ASC卷积核调制的SAR目标识别深层网络在实现更加简单的情况下识别性能依然具有优势。此外，对表4所提方法与对比方法的参数量和性能进行对比可以发现，与VGG16和ResNet34相比，所提方法不仅大幅降低了参数量，且显著提升了识别性能；与A-ConvNet和BaseNet相比，所提方法能够在参数量相当的情况下显著提升识别性能；而对于CA-MCNN来说，表4给出的只是网络的参数量，但该方法还需要对每张SAR图像提取ASC，与其相比，本文所提方法能够在性能相当的情况下，大大降低参数量。综上所述，通过结合SAR目标的电磁散射特性，相比对

比方法，所提方法能够在参数量相当的情况下大幅提升识别性能，或者在识别性能相当的情况下大幅降低参数量。

基于3类7型MSTAR子数据集的实验结果证明了本文所设计的网络的有效性，其可以通过使用ASC调制卷积核提取更符合SAR目标特性的散射结构和边缘特征，从而在降低网络参数数量的同时保证网络的特征提取能力，进一步降低了对训练样本的需求。

4.3.2 基于10类14型MSTAR子数据集的识别实验

本节将进一步讨论所设计的基于ASC卷积核调制的SAR目标识别深层网络对10类14型MSTAR子数据集的有效性，表5展示了不同方法对10类14型MSTAR子数据集的识别性能以及各方法的参数量。

首先，对比表4和表5可以发现，各方法在10类14型MSTAR子数据集上的识别性能相比在3类7型MSTAR子数据集上的性能都有所下降。这是因为，对于10类14型MSTAR子数据集来说，其测试数据中不仅包含未在训练数据中出现过的变体型号，其还包含额外的7类军事目标，且每一类军事目标的训练样本都是有限的，因此，相对于3类7型MSTAR子数据集，10类14型MSTAR子数据集的识别难度更高，从而导致各方法在该子数据集上的识别性能有所下降。其次，从表5可以看出，所提方法的PCC可以达到0.9844。与表4的结果类似，所提方法的PCC不仅高于VGG16和ResNet34这两种参数量高的SAR目标识别方法，也高于A-ConvNet这种参数量低的SAR目标识别方法，同时其比BaseNet和CA-MCNN的PCC值也要高。导致该结果的原因依然是本文设计的网络使用了ASC调制卷积核，其不仅能够在只学习一小组CNN卷积核的情况下，通过ASC调制得到更多的卷积核，而且能够提取更符合SAR图像特性的散射结构和边缘特

表4 不同方法对3类7型MSTAR子数据集的识别性能
以及各方法参数量

Tab. 4 Recognition performance of different methods on three-target MSTAR data and the number of parameters of each method

方法	参数量	PCC
VGG16	6.51×10^7	0.9319
ResNet34	2.13×10^7	0.9253
A-ConvNet	3.03×10^5	0.9385
BaseNet	2.27×10^6	0.9495
CA-MCNN	6.52×10^6	0.9861
所提方法	2.26×10^6	0.9875

表5 不同方法对10类14型MSTAR子数据集的识别性能
以及各方法参数量

Tab. 5 Recognition performance of different methods on ten-target MSTAR data and the number of parameters of each method

方法	参数量	PCC
VGG16	6.51×10^7	0.9166
ResNet34	2.13×10^7	0.9138
A-ConvNet	3.03×10^5	0.9219
BaseNet	2.27×10^6	0.9422
CA-MCNN	6.52×10^6	0.9781
所提方法	2.26×10^6	0.9844

征,因此本文设计的网络可以在降低参数数量的同时保证CNN的特征提取能力,从而在训练数据有限的10类14型MSTAR子数据集上取得更好的识别性能。

此外,对表5所提方法与对比方法的参数数量和性能进行仔细对比可以得到与表4相同的结论,通过结合SAR目标的电磁散射特性,相比对比方法,所提方法能够在参数数量相当的情况下大幅提升识别性能,或者在识别性能相当的情况下大幅降低参数数量。

基于10类14型MSTAR子数据集的实验结果进一步证明了本文所设计的网络的有效性,其可以通过使用ASC调制卷积核提取更符合SAR目标特性的散射结构和边缘特征,从而在降低网络参数数量的同时保证网络的特征提取能力,即便在面对包含更多目标种类的难度更大的数据集时也能取得良好的识别性能。

4.3.3 基于OpenSARShip舰船数据集的识别实验

本节将进一步讨论所设计的基于ASC卷积核调制的SAR目标识别深层网络对OpenSARShip舰船数据集的有效性。由于CA-MCNN方法需要进行ASC提取,需要使用具有真实雷达分辨率和带宽参数的复数据,而OpenSARShip舰船数据集的两类图像产品均不能满足,因此本节不使用CA-MCNN作为对比方法。表6展示了不同方法对3类OpenSARShip舰船数据集目标的识别性能以及各方法的参数数量。

首先,与MSTAR数据集上的识别效果对比可见,各方法在OpenSARShip舰船数据集3类目标上的识别性能相比MSTAR数据集都有所下降,这是由于OpenSARShip舰船数据集具有更复杂的背景,且舰船目标本身类内差异大、类间差异相对小,因此OpenSARShip舰船数据集的识别难度更高,从而导致各方法的识别性能的下降。此外,如表6所示,本文设计的网络对OpenSARShip 3类舰船数据集的PCC可以达到0.8101,优于对比方法。对于VGG16和ResNet34来说,OpenSARShip舰船数据集3类目标的数据量更小,难以完全训练网络中大量的参数,因此VGG16和ResNet34识别效果不佳。对于A-ConvNet,其少量的参数限制了特征提取能力,因此对于更难分的舰船目标识别性能也相对较差。最后,与BaseNet相比,所提方法依然能取得更高的PCC,说明所提方法所使用的ASC调制卷积核相比于普通卷积层可以提取更适用于SAR图像的特征,证明了所提方法的有效性。因此所提方法仍能在背景更为复杂的OpenSARShip舰船数据

集上取得最佳的识别性能,说明所提方法对于复杂背景下的SAR目标识别具有鲁棒性。

4.4 模型分析

4.4.1 预设的ASC核的方向数对识别性能的影响

3.2节分析了分布式ASC的指向 φ 的取值,为了从SAR图像中提取不同方向的散射结构和边缘特征,所提方法令 φ 在 $[-90^\circ, 90^\circ]$ 之间均匀取值,假设共取 M 个值,则可以得到 M 个不同指向的分布式ASC。考虑到 φ 的取值越多,得到的分布式ASC的指向越丰富,但同时也会增加网络的参数数量,因此所提方法综合考虑参数数量和指向丰富性,令 $M=4$,即 $\varphi = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$ 。为了进一步分析预设的ASC核的方向数对识别性能的影响,本节基于3类7型MSTAR子数据集,在表7给出了当分布式ASC的指向 φ 取不同的 M 值时的识别性能。

从表7可以看出,预设的ASC核的方向数取不同的 M 值时会对识别性能产生影响,当指向数 M 太小时,如 $M=2$ 时,预设的ASC核的指向不够丰富,无法提取不同方向的散射结构和边缘特征,从而会导致识别性能下降;当指向数 M 太大时,如 $M=6$ 时,虽然保证了预设的ASC核的丰富性,但也增加了网络的参数数量,因此在训练样本有限时,也会导致识别性能的下降。如表7所示,当 $M=4$ 时,本文所设计的网络在3类7型MSTAR子数据集上取得了最高的识别性能,这是因为当 $M=4$ 时, $\varphi = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$,既保证了预设的ASC核的指向丰富性,同时又不会导致参数数量过多。综上所述,表7证明了所提方法将预设的ASC核的指向数 M 设置为4的合理性。

4.4.2 预设的ASC核的长度值对识别性能的影响

3.2节分析了分布式ASC的长度 L 的取值, L 的取值由图像的空间分辨率以及CNN的卷积核尺寸

表6 不同方法对OpenSARShip的3类数据集的识别性能
以及各方法参数数量

Tab. 6 Recognition performance of different methods on three-target OpenSARShip data and the number of parameters of each method

方法	参数数量	PCC
VGG16	6.51×10^7	0.7713
ResNet34	2.13×10^7	0.7403
A-ConvNet	3.03×10^5	0.7791
BaseNet	2.27×10^6	0.8062
CA-MCNN	6.52×10^6	—
所提方法	2.26×10^6	0.8101

共同决定，由于MSTAR数据集的空间分辨率为 $0.3\text{ m} \times 0.3\text{ m}$ ，因此，对于 5×5 的卷积核， L 的取值可以为 $[0.3, 0.6, 0.9, 1.2]$ 。此外，由于所提方法希望通过ASC调制核提取更符合SAR图像特性的散射结构和边缘特征，因此预设的ASC核中的分布式ASC的长度不应太短，同时又为了保证ASC核中分布式ASC的完整性，结合图2所示的ASC核可视化结果，最终选择 $L = 0.9$ 时对应的ASC核作为预设的一组ASC核。为了进一步分析预设的ASC核的长度值对识别性能的影响，本节基于3类7型MSTAR子数据集，在表8中给出了当分布式ASC的长度 L 取不同值时的识别性能。

从表8可以看出，预设的ASC核的长度 L 取不同值时会对识别性能产生影响，当长度 L 太小时，如 $L = 0.3$ 时，无法有效提取SAR图像中的散射结构和边缘特征，从而会导致识别性能下降；而当长度 L 取1.2时，如图2的ASC核可视化结果所示，无法保证ASC核中分布式ASC的完整性，因此也会对识别性能造成影响。如表8所示，当 $L = 0.9$ 时，本文所设计的网络在3类7型MSTAR子数据集上取得了最高的识别性能，这是因为当 $L = 0.9$ 时，既能有效提取更符合SAR图像特性的散射结构和边缘特征，又能保证ASC核中分布式ASC的完整性。综上所述，表8证明了所提方法将预设的ASC核的长度 L 设置为0.9的合理性。

表 7 预设的ASC核的方向数取不同的 M 值时对3类7型

MSTAR子数据集的识别性能

Tab. 7 Recognition performance on three-target MSTAR data under different values of M

方向数 M	PCC
2	0.9722
3	0.9817
4	0.9875
5	0.9810
6	0.9744

表 8 预设的ASC核的长度 L 取不同值时对3类7型

MSTAR子数据集的识别性能

Tab. 8 Recognition performance on three-target MSTAR data under different values of L

长度 L	PCC
0.3	0.9729
0.6	0.9788
0.9	0.9875
1.2	0.9810

4.4.3 不同层卷积层采用ASC调制卷积核对识别性能的影响

本文设计的网络共包含5个卷积层和2个全连接层，考虑到CNN的浅层负责提取图像中的散射结构和边缘特征，所提方法在前2个卷积层使用本文设计的ASC调制卷积核，以提取更符合SAR图像特性的散射结构和边缘特征。后3个卷积层依然使用CNN卷积核，以保证提取更具鉴别性的语义特征。所提ASC调制卷积核是一个即插即用的模块，可以直接用于替换VGG, ResNet等深度网络的卷积层。为了进一步分析不同网络结构下不同层卷积层采用ASC调制卷积核对识别性能的影响，本节基于3类7型MSTAR子数据集，在表9中给出了VGG16, ResNet18与所提方法替换不同层的卷积核为ASC调制卷积核时的识别性能。

首先，从表9可以看出，采用ASC调制卷积核可以提升3个网络的识别效果，说明ASC调制卷积核可以作为一个即插即用的模块，用于提升深度网络的SAR目标识别效果。其次，在共两层卷积层采用ASC调制卷积核的情况下，采用的层数越浅，网络的识别性能越好。浅层卷积层采用ASC调制卷积核可以为浅层特征引入SAR图像的电磁特性，更适用于SAR目标的识别，因此应当从网络的第1层卷积层开始采用ASC调制卷积层。最后，从第1层开始，采用ASC调制卷积核的卷积层层数不同会对识别性能产生影响。当只有第1个卷积层采用ASC调制卷积核时，3个网络对SAR图像中的散射结构和边缘特征提取得不够充分，而当前4个卷积层都采用ASC调制卷积核时又会影响整个网络对SAR图像的语义特征的提取能力，这些情况都会对识别性能产生负面影响。如表9所示，当前两个卷积层采用ASC调制卷积核时，本文所设计的网络在3类7型MSTAR子数据集上取得了最高的识别性能。

表 9 不同层卷积层采用ASC调制卷积核时对3类7型

MSTAR子数据集的识别性能

Tab. 9 Recognition performance on three-target MSTAR data under different convolution layers using ASC modulated convolutional kernels

采用层	VGG16	ResNet18	所提方法
不采用	0.9319	0.9480	0.9495
第1层	0.9795	0.9663	0.9795
第1, 2层	0.9839	0.9769	0.9875
第1~3层	0.9773	0.9736	0.9810
第1~4层	0.9758	0.9729	0.9736
第2, 3层	0.9766	0.9641	0.9751
第3, 4层	0.9714	0.9582	0.9707

综上所述, ASC调制卷积核作为即插即用的模块时, 应当从第一层开始采用ASC卷积核, 具体采用层数应当根据具体网络结构以及训练数据进行选择, 本文的实验结果可以作为选择的参考。

4.4.4 不同训练样本数对识别性能的影响

本文设计的网络使用了ASC调制卷积核, 该卷积核能够提取更符合SAR图像特性的散射结构和边缘特征。具体来说, ASC调制卷积核是用预先设定好的一组ASC核对CNN的可学习卷积核进行调制得到的, 这样可以在只学习一小组CNN卷积核的情况下, 通过ASC调制得到更多的卷积核, 从而可以在降低参数量的同时保证CNN的特征提取能力, 进一步降低对训练样本的需求量。为了验证本文所设计的网络可以在降低对训练样本需求的同时保证优秀的识别性能, 本节基于3类7型MSTAR子数据集, 在表10中给出了不同方法在不同训练样本数下的识别性能。

如表10所示, 本节探索了当训练样本取100%, 50%, 30%, 25%和20%时, 不同SAR目标识别方法的性能。首先, 对于VGG16和ResNet34来说, 由于其参数量很大, 其对训练样本的需求也很大, 因此, 随着训练样本数的减少, 这两种方法的性能下降非常明显。其次, 对于A-ConvNet和BaseNet来说, 其参数量更少, 因此随着训练样本数的减少, 这两种方法的性能要高于VGG16和ResNet34, 但由于A-ConvNet和BaseNet只使用了SAR图像信息, 其识别性能依然需要依赖训练样本的完备性, 所以随着训练样本数的减少, 这两种方法的性能也受到了明显的影响。与上述4种方法相比, 随着训练样本数的减少, CA-MCNN依然能够保证优秀的识别性能, 这是因为该方法能够利用SAR目标的图像信息和ASC信息来学习描述能力更强的特征, 从

而提升训练样本有限的情况下的目标识别性能。与CA-MCNN相比, 本文设计的网络在训练样本数减少时能够取得相对更好的性能, 具体来说, 在训练样本只取25%时, 所提方法的PCC依然能够达到0.9187。这是因为所提方法使用了ASC调制卷积核, 其不仅能够降低网络的参数量, 同时能够提取更符合SAR图像特性的散射结构和边缘特征, 因此在训练样本不足的情况下依然能够取得优秀的识别性能。综上所述, 表10证明了本文所设计的基于ASC卷积核调制的SAR目标识别深层网络的有效性, 其可以在降低对训练样本需求的同时保证优秀的目标识别性能。

5 结语

本文研究了基于ASC卷积核调制的SAR目标识别深层网络, 通过结合SAR目标的电磁散射特性设计了更简单的网络结构来提取更符合SAR图像特性的特征, 从而在降低网络参数量的同时保证网络的特征提取能力, 实现了在降低对训练样本需求的同时保证优秀的目标识别性能的目的。首先, 所提方法设计了ASC调制卷积核, 其使用预先设定的具有不同指向和长度的ASC核对CNN本身的卷积核进行调制, 从而利用少量可学习卷积核生成更多的卷积核。其次, 所提方法仅在网络的浅层使用ASC调制卷积核, 用以提取更符合SAR图像特性的散射结构和边缘特征, 而在网络的高层使用CNN本身的卷积核, 用以提取SAR图像的语义特征。由于整个网络同时使用了ASC调制卷积核和CNN本身的卷积核, 因此该网络能够兼顾SAR目标的电磁散射特性和CNN的特征提取优势。基于MSTAR实测数据集和OpenSARShip舰船数据集的实验结果表明, 本文设计的网络可以在降低对训练样本需求的同时保证优秀的目标识别性能。

表 10 不同方法在不同训练样本数下对3类7型MSTAR子数据集的识别性能

Tab. 10 Recognition performance of different methods on three-target MSTAR data with different number of training samples

方法	训练样本比例				
	100%	50%	30%	25%	20%
VGG16	0.9319	0.8886	0.8352	0.7883	0.7370
ResNet34	0.9253	0.8813	0.8278	0.7590	0.7267
A-ConvNet	0.9385	0.9062	0.8769	0.8337	0.7875
BaseNet	0.9495	0.8974	0.8674	0.8227	0.7758
CA-MCNN	0.9861	0.9641	0.9165	0.8938	0.8608
所提方法	0.9875	0.9670	0.9480	0.9187	0.8711

利益冲突 所有作者均声明不存在利益冲突

Conflict of Interests The authors declare that there is no conflict of interests

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Deep Network for SAR Target Recognition Based on Attribute Scattering Center Convolutional Kernel Modulation

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Abstract: The feature extraction capability of Convolutional Neural Networks (CNNs) is related to the number of their parameters. Generally, using a large number of parameters leads to improved feature extraction capability of CNNs. However, a considerable amount of training data is required to effectively learn these parameters. In practical applications, Synthetic Aperture Radar (SAR) images available for model training are often limited. Reducing the number of parameters in a CNN can decrease the demand for training samples, but the feature expression ability of the CNN is simultaneously diminished, which affects its target recognition performance. To solve this problem, this paper proposes a deep network for SAR target recognition based on Attribute Scattering Center (ASC) convolutional kernel modulation. Given the electromagnetic scattering characteristics of SAR images, the proposed network extracts scattering structures and edge features that are more consistent with the characteristics of SAR targets by modulating a small number of CNN convolutional kernels using predefined ASC kernels with different orientations and lengths. This approach generates additional convolutional kernels, which can reduce the network parameters while ensuring feature extraction capability. In addition, the designed network uses ASC-modulated convolutional kernels at shallow layers to extract scattering structures and edge features that are more consistent with the characteristics of SAR images while utilizing CNN convolutional kernels at deeper layers to extract semantic features of SAR images. The proposed network focuses on the electromagnetic scattering characteristics of SAR targets and shows the feature extraction advantages of CNNs due to the simultaneous use of ASC-modulated and CNN convolutional kernels. Experiments based on the studied SAR images demonstrate that the proposed network can ensure excellent SAR target recognition performance while reducing the demand for training samples.

Key words: Synthetic Aperture Radar (SAR); Target recognition; Convolutional Neural Network (CNN); Attribute Scattering Center (ASC); Convolutional kernel modulation

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1 Introduction

A synthetic Aperture Radar (SAR) is an active microwave sensor that achieves high-resolution 2D imaging using a large-aperture antenna

synthesized through the movement of a small physical antenna along a straight trajectory. SAR can provide all-weather, wide-area, long-range, high-resolution imaging of the Earth's surface without being affected by lighting or atmospheric conditions such as clouds and fog. It also exhibits certain penetration capabilities. Owing to these advantages, SAR technology has played an increasingly important role in civilian and military applications, including national defense, agricultural production estimation, and geological surveying. As one of the most critical steps in SAR image interpretation, SAR target recognition aims to classify input target regions into predefined

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categories. This process holds immeasurable application value in various civilian and military fields and has attracted extensive worldwide research attention, yielding abundant research achievements.

With the emergence of large-scale datasets and the growth of computational power, deep neural networks have been extensively studied and applied in target recognition. To simulate the cognitive mechanisms of the human brain, deep networks employ multilayered model structures to achieve multilevel nonlinear transformations. Raw data are mapped into feature spaces suitable for recognizing and improving target recognition performance. As a typical deep network for image classification, Convolutional Neural Networks (CNNs)^[1-6] integrate feature extraction and classification within a unified framework and demonstrate powerful feature learning capabilities. The deep features learned through convolutional operations exhibit strong discriminative power for different types of targets^[1]. CNNs have become a mainstream approach in SAR target recognition. Chen *et al.*^[7] constructed a SAR target recognition network called A-ConvNet that contains only convolutional layers; they reduced the number of network parameters by removing fully connected layers, alleviating the overfitting problem caused by insufficient training data to some extent. Yu *et al.*^[8] proposed a SAR target recognition method that is based on fully convolutional networks and improved convolutional autoencoders and demonstrates strong noise resistance. Zhao *et al.*^[9] utilized the spatial features of SAR targets extracted by EfficientNet^[4] and further employed recurrent neural networks to extract temporal features from image sequences, thereby achieving robust SAR target recognition. Huang *et al.*^[10] proposed a lightweight CNN for SAR target recognition that incorporates global and local branches to combine multilevel features for target identification. Liu *et al.*^[11] designed a complex-valued convolutional network for SAR image recognition and improved its recognition performance by incorporating the electromagnetic scattering characteristics of SAR images into the initialization of shal-

low convolutional layers. Zhang *et al.*^[12] fused traditional handcrafted HOG features with deep network features to design a recognition network suitable for ship targets in SAR images. Qosja *et al.*^[13] applied ConvNeXt^[14] to SAR target recognition and verified the applicability of certain data augmentation methods for SAR data. Zhang *et al.*^[15] constructed a dual-branch CNN that extracts visual features and topological features for few-shot ship classification.

The aforementioned methods of CNN-based SAR target recognition can automatically extract classification-appropriate features from training data, thus avoiding the shortcomings of traditional manual feature extraction that relies heavily on human expertise. However, the feature extraction capability of CNNs is closely related to their parameter quantity. In general, networks with many parameters demonstrate strong feature extraction capabilities but require large amounts of training data for parameter optimization. In practice, the acquisition of high-quality SAR images entails expensive equipment and resources, making it remarkably more challenging compared with obtaining optical images. In addition, the coherent imaging mechanism of SAR creates several inherent problems, including geometric distortions, structural loss of targets in SAR images, and sensitivity to observation parameters. These factors complicate the establishment of comprehensive SAR target recognition datasets. In practical applications, the available training samples for SAR target recognition are often limited in quantity. This scarcity frequently leads to severe overfitting during network training, which negatively impacts the network's feature extraction capability and ultimately degrades recognition performance. Two main approaches have been explored to address this issue. On the one hand, data augmentation can be employed. However, most existing methods for SAR target recognition directly apply data augmentation approaches developed for optical images to SAR data; the applicability of SAR data augmented using these approaches cannot be guaranteed. This practice may also generate augmented res-

ults that violate SAR image characteristics, potentially impairing target recognition performance. On the other hand, reducing the number of CNN parameters can decrease the demand for training samples but simultaneously weaken the network's feature representation capacity, consequently affecting its target recognition performance. Therefore, a deep network architecture specific for SAR target recognition that can effectively extract SAR image features under training with limited data must be designed. Such a network should maintain robust feature representation capabilities while being less prone to overfitting when trained with small datasets, thereby addressing the fundamental challenges in SAR target recognition.

To intuitively understand what CNNs learn from images, researchers^[16] took AlexNet^[1] trained on the ImageNet dataset as an example and visualized the first-layer convolutional kernels. They demonstrated that these kernels primarily extract linear structures and edge features. However, SAR images are obtained on the basis of the principle of electromagnetic backscattering. Compared with optical images, they exhibit discrete and discontinuous characteristics^[17]. Their salient structures primarily comprise scattering patterns and edge features. Therefore, the electromagnetic scattering properties of SAR images must be considered when extracting their scattering structures and edge features. The Attributed Scattering Center (ASC) model^[18,19] employs seven parameters to describe the characteristics of each scattering center, encompassing rich physical and geometric attributes that precisely characterize the true scattering mechanisms of targets. This model can accurately represent the scattering properties of targets in high-resolution SAR images. In this work, a deep network based on ASC-modulated convolutional kernels for SAR target recognition is proposed to design a simple network structure that incorporates the electromagnetic scattering properties of SAR targets to extract features aligned with SAR image characteristics. This approach reduces the number of network parameters while maintaining strong

feature extraction capability. The proposed method employs ASC-modulated convolutional kernels in the shallow layers of the network to extract scattering structures and edge features that match SAR image properties. Standard CNN convolutional kernels are used in deeper layers to extract semantic features. ASC-modulated kernels are generated by modulating the CNN's original kernels using predefined ASC kernels of varying orientations and lengths. This operation allows a small number of learnable kernels to generate a large set of effective filters, thereby reducing the network's parameter count while preserving its feature extraction capacity. Given that the network combines ASC-modulated and standard CNN kernels, it can leverage the electromagnetic scattering properties of SAR targets and the powerful feature learning ability of CNNs and thus further reduce the demand for training samples. Experimental results on the MSTAR and OpenSARShip datasets demonstrate that the proposed network achieves superior target recognition performance while requiring few training samples.

2 ASC Model

SAR images are obtained on the basis of the principle of electromagnetic backscattering. Thus, their electromagnetic scattering characteristics are distinct from those of optical images. When a radar operates in the high-frequency band, the scattered field from distributed targets can be approximated as the superposition of the scattered fields from individual scattering centers that constitute the target^[18]. The ASC model is a classical scattering center model that describes the dependence of SAR target backscattering on frequency and azimuth. Its parameters contain rich physical and geometric attributes that enable an accurate characterization of the electromagnetic scattering properties of radar targets in the high-frequency band. Hence, this model has garnered extensive attention and wide applications. For distributed targets in the high-frequency band, their scattered field can be expressed as

$$E(f, \varphi; \Theta) = \sum_{i=1}^K E_i(f, \varphi; \theta_i) \quad (1)$$

where f denotes the radar frequency, φ represents the azimuth angle, Θ corresponds to the parameter set of ASCs, and K indicates the total number of ASCs within the target. For an individual ASC, its scattered field can be mathematically represented by the ASC model in Eq. (2)^[18]:

$$E_i(f, \varphi; \theta_i) = A_i \cdot \left(j \frac{f}{f_c} \right)^{\alpha_i} \cdot \exp \left(\frac{-j4\pi f}{c} (x_i \cos \varphi + y_i \sin \varphi) \right) \cdot \text{sinc} \left(\frac{2\pi f}{c} L_i \sin(\varphi - \bar{\varphi}_i) \right) \cdot \exp(-2\pi f \gamma_i \sin \varphi) \quad (2)$$

where f_c represents the radar center frequency, j denotes the imaginary unit, c corresponds to the propagation speed of electromagnetic waves (*i.e.*, the speed of light), and $\text{sinc}(\cdot) = \sin(\cdot)/(\cdot)$ indicates the sinc function. The attribute parameters of the i -th ASC can be expressed as $\theta_i = [A_i, \alpha_i, x_i, y_i, L_i, \bar{\varphi}_i, \gamma_i]$ ($i = 1, 2, \dots, K$), where A_i represents the backscattering coefficient (a complex value); α_i denotes the frequency dependence factor, describing the dependence of the ASC's backscattering amplitude on frequency; x_i and y_i represent the position coordinates of the ASC in range and azimuth dimensions, respectively; L_i indicates the length of the distributed ASC; $\bar{\varphi}_i$ shows the orientation of the distributed ASC; and γ_i stands for the azimuth dependence factor, characterizing the dependence of the ASC's backscattering amplitude on azimuth angle. If the i -th ASC is a localized ASC, then it satisfies $L_i =$

$\bar{\varphi}_i = 0$; If the i -th ASC is a distributed ASC, then it satisfies $\gamma_i = 0$ and $L_i > 0$.

3 Methods

3.1 Network architecture and process

Fig. 1 illustrates the architecture of the proposed deep network based on ASC convolutional kernel modulation for SAR target recognition. As shown in Fig. 1, the network consists of five convolutional layers and two fully connected layers. Given that SAR images are acquired through electromagnetic backscattering and exhibit scattering characteristics distinct from those of optical images, feature extraction should incorporate these electromagnetic scattering properties to obtain SAR-specific representations. As shallow CNN layers primarily extract low-level structural features, our method employs the proposed ASC-modulated convolutional kernels in the first two convolutional layers to capture the scattering structures and edge features consistent with SAR image properties. The remaining three convolutional layers contain standard CNN kernels to ensure the extraction of discriminative semantic features. After passing through five convolutional layers, the extracted features are processed through two fully connected layers to match the dimension of target categories and then through a softmax layer for final classification. The entire network is end-to-end trained using cross-entropy loss with Stochastic Gradient Descent (SGD) for parameter optimization.

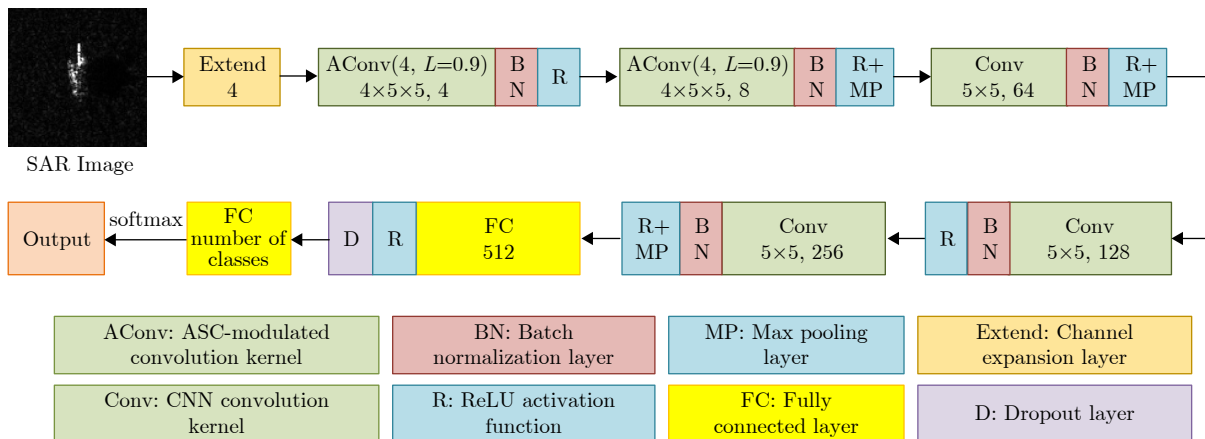


Fig. 1 Architecture of the deep network for SAR target recognition based on ASC convolutional kernel modulation

The following sections systematically elucidate the generation process, convolutional operations, and parameter updating mechanism of the ASC-modulated convolutional kernels in the proposed deep learning network for SAR target recognition.

3.2 Generation of ASC-modulated convolutional kernels

SAR images are acquired through electromagnetic backscattering and exhibit electromagnetic scattering characteristics distinct from those of optical images. When the radar operates in the high-frequency band, the scattered field from distributed targets can be approximated as the coherent summation of scattered fields from individual constituent scattering centers^[18]. To extract scattering structures and edge features representing SAR image characteristics, this work proposes ASC-modulated convolutional kernels generated by modulating the learnable CNN kernels with a predefined set of ASC kernels. This approach enables the generation of a large set of effective kernels through the learning of only a small group of base CNN kernels, thereby reducing the number of parameters and preserving the network's feature extraction capability.

A set of ASC kernels must first be pregenerated to obtain the ASC-modulated convolution kernel. As discussed in Section 2, each ASC contains seven attribute parameters denoted as $\theta = [A, \alpha, x, y, L, \bar{\varphi}, \gamma]$. For the generation of ASC kernels, each ASC is first positioned at the kernel center by setting $x = y = 0$, and the ASC amplitude is then normalized by setting $A = 1$. Given that the proposed method aims to extract scattering structures and edge features suitable for SAR images using ASC, this study employs distributed ASCs, setting $\gamma = 0, L > 0$. The parameters of the ASC model were analyzed in Ref. [20]. For high-resolution SAR images where the bandwidth is typically smaller than the carrier frequency, α has minimal influence on the magnitude of the target backscattering field and thus can be approximated as zero. Therefore, only L and $\bar{\varphi}$ need to be determined, where L represents the length of the distributed ASC, and $\bar{\varphi}$ denotes its orienta-

tion angle. $\bar{\varphi}$ is set to take uniformly spaced values within the range $[-90^\circ, 90^\circ]$ to extract scattering structures and edge features from SAR images in different orientations. Assuming that a total of M values are selected, M -distributed ASCs with distinct orientations can be obtained. Meanwhile, L is determined from the image's spatial resolution and the CNN's kernel size. Spatial resolution refers to the actual distance represented by two adjacent pixels in the input image. Assuming that the spatial resolution is m and the convolution kernel size is $r \times r$, then $L = p \times m$ ($p = 1, 2, \dots, r - 1$).

In CNNs, a large convolution kernel size corresponds to a large receptive field. For SAR images, an expanded receptive field can mitigate the impact of speckle noise to some extent. However, increasing the kernel size also increases the number of trainable parameters in the network. In the proposed network, all convolutional layers employ 5×5 kernels. The MSTAR dataset^[21] with spatial resolution of $0.3 \text{ m} \times 0.3 \text{ m}$ is taken as an example. For a 5×5 convolution kernel, L can be set to $[0.3, 0.6, 0.9, 1.2]$. Although a large number of $\bar{\varphi}$ values would yield diverse orientations for the distributed ASCs, it would also increase the network's parameter count. In this work, $M = 4$, *i.e.*, $\bar{\varphi} = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$, to balance parameter efficiency and orientation diversity. A set of 5×5 ASC kernels is generated by combining L and $\bar{\varphi}$, as illustrated in Fig. 2. Given that the proposed method aims to extract scattering structures and edge features aligned with SAR image characteristics using ASC-modulated kernels, the predefined distributed ASCs should not be too short in length. Moreover, the final selection—based on the visualization results in Fig. 2—adopts the ASC kernels corresponding to $L = 0.9$ and $\bar{\varphi} = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$ (marked by the red box in Fig. 2) to ensure the integrity of the distributed ASCs within the kernels.

Given that the selected ASC kernels consist of a set of kernels with different orientations, the proposed method employs 3D CNN convolution kernels to encode directional information into the ASC-modulated kernels. The size of a 3D CNN

convolution kernel can be expressed as $N \times W \times W$, where $W \times W$ denotes the spatial dimensions of the kernel and represents the number of channels. On the basis of the preceding analysis, 3D CNN kernels with $W = 5$ and $N = 4$ are adopted in this study. An ASC-modulated convolution kernel is generated by modulating a learnable 3D CNN kernel with the predefined set of ASC kernels exhibiting varying orientations (highlighted by the red box in Fig. 2). The modulation process, which applies the selected ASC kernels to the 3D CNN kernel, is illustrated in Fig. 3 and mathematically formulated in Eq. (3).

$$\mathbf{AC}_{i,m} = \mathbf{C}_i \circ \mathbf{A}(m) \quad (3)$$

where $\mathbf{AC}_{i,m}$ denotes the i -th directional channel of the m -th ASC-modulated convolution kernel, where $m = 1, 2, 3, 4$; $\mathbf{C}_i$ represents the i -th learnable 3D CNN convolution kernel; and $\mathbf{A}(m)$ is

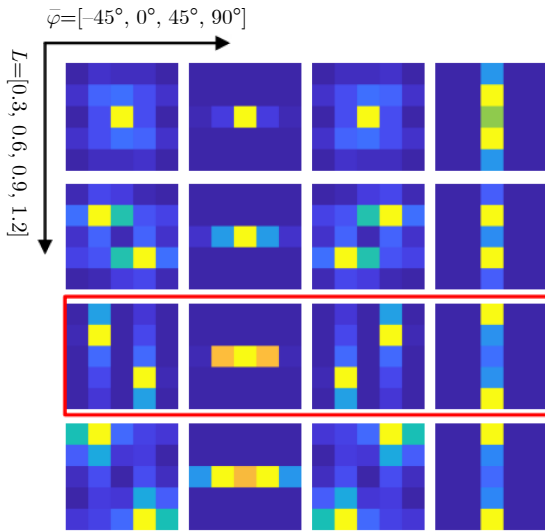


Fig. 2 A set of ASC kernels obtained according to the combinations of L and $\bar{\varphi}$ (the ASC kernels within the red box are the final selected ASC kernels)

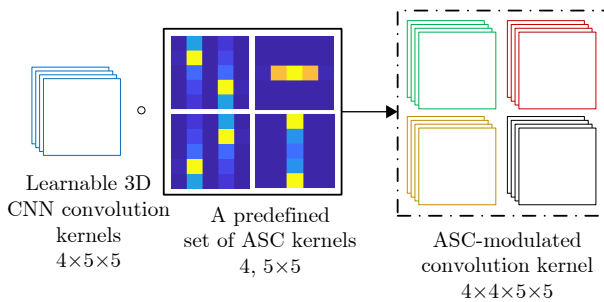


Fig. 3 The generation process of ASC-modulated convolutional kernel

the m -th ASC kernel from the predefined set of ASC kernels with varying orientations (obtained by taking the modulus of the complex-valued ASC kernels to yield real-valued ASC kernels) and mathematically expressed in Eq. (2). The symbol $\circ$ denotes element-wise multiplication between the corresponding channels of $\mathbf{A}(m)$ and $\mathbf{C}_i$. Thus, the i -th ASC-modulated convolution kernel can be expressed as

$$\mathbf{AC}_i = (\mathbf{AC}_{i,1}, \mathbf{AC}_{i,2}, \mathbf{AC}_{i,3}, \mathbf{AC}_{i,4}) \quad (4)$$

As shown in Eq. (4), the i -th ASC-modulated convolution kernel $\mathbf{AC}_i$ is essentially composed of four 3D convolution kernels illustrated in Fig. 3. For each ASC-modulated convolution kernel, its dimensions can be expressed as $U \times N \times W \times W$, where U represents the number of directional channels. The resulting ASC-modulated kernels have dimensions of $U = 4, N = 4, W = 5$. Given that the ASC kernels are predefined, only the parameters of the learnable 3D CNN kernel $N \times W \times W$ need to be updated during training. Therefore, this modulation enables the proposed method to extract discriminative features suited for SAR targets without increasing the number of parameters. Moreover, the predefined ASC kernels inherently encode multiple orientations, allowing the trained ASC-modulated convolution kernels to automatically extract target features under different aspect angles and thereby enhancing recognition performance.

3.3 Convolution of ASC-modulated convolutional kernels

In the proposed network architecture, ASC-modulated convolution kernels are employed to generate feature maps. Compared with the feature maps produced by conventional CNN convolution kernels, the ASC-modulated kernels exhibit superior capability in extracting scattering structures and edge features aligned with SAR image characteristics. Given that each ASC-modulated convolution kernel consists of four 3D convolution kernels, a single-channel, 2D input image must first undergo channel expansion to transform into a tensor of dimensions $1 \times 4 \times S_i \times S_i$. Here, the original single channel (denoted as 1) is expanded to four channels to en-

code orientation information. These expanded channels are referred to as orientation channels, and S_i represents the spatial dimensions of the input image. For an input image, the feature map generated by an ASC-modulated convolution kernel $\hat{\mathbf{F}}$ can be mathematically expressed as

$$\hat{\mathbf{F}} = \text{AConv}(\mathbf{F}, \mathbf{AC}_i) \quad (5)$$

where $\text{AConv}(\cdot)$ denotes the ASC-modulated convolution operation, $\mathbf{F}$ represents the input image, and $\mathbf{AC}_i$ denotes the i -th ASC-modulated convolution kernel. The convolution process described by Eq. (5) is illustrated in Fig. 4, where S_o indicates the dimensions of the output image. The k -th channel of the output feature map $\hat{\mathbf{F}}$ can be expressed as

$$\hat{\mathbf{F}}_k = \sum_{n=1}^4 \mathbf{F}^{(n)} * \mathbf{AC}_{i,m}^{(n)} \quad (6)$$

where $\mathbf{F}^{(n)}$ represents the n -th directional channel of the output feature map $\mathbf{F}$, $\mathbf{AC}_{i,m}^{(n)}$ denotes the n -th kernel in the m -th directional channel of the i -th ASC-modulated convolution kernel, and $*$ indicates the convolution operation. According to Eq. (6), each channel of the output feature map $\hat{\mathbf{F}}$ corresponds to a distinct orientation in the predefined ASC kernels. Through this convolutional operation, the proposed method effectively encodes orientational information into the feature maps.

In the network architecture, the input to each convolutional layer typically consists of multichannel feature maps. Hence, the complete convolution process for a layer employing ASC-modulated convolution kernels can be described as follows: consider a convolutional layer with input feature maps of dimensions $5 \times 4 \times S_i \times S_i$,

where 5 represents the number of input feature map channels, 4 denotes the number of orientation channels used for directional information encoding, and S_i indicates the spatial dimensions of the input feature maps. When this layer employs 10 sets of ASC-modulated convolution kernels, the kernel dimensions are changed to $10 \times 5 \times 4 \times 4 \times 5 \times 5$ to maintain channel consistency with the input; here, 10 corresponds to the number of kernel groups, 5 matches the input channel dimension, and $4 \times 4 \times 5 \times 5$ specifies the spatial size of each ASC-modulated kernel. Owing to the modulation effect of the predefined ASC kernels, the learnable parameters in this convolutional layer are reduced to dimensions $10 \times 5 \times (4 \times 5 \times 5 + 1)$, amounting to only one-fourth of the parameter count required in conventional layers. The complete convolution process for this layer is illustrated in Fig. 5, where S_o represents the spatial dimensions of the output.

3.4 Parameter update of ASC-modulated convolutional kernels

Although the ASC-modulated convolution kernels are obtained by modulating CNN convolution kernels using ASC kernels, only the parameters of the learnable CNN kernels $\mathbf{C}_i$ require updates during backpropagation because the ASC kernels are predetermined. Similar to the update rule for conventional CNN convolution kernels, the parameter update for ASC-modulated convolution kernels also employs SGD and the chain rule. The formula for parameter update is as follows:

$$\delta = \frac{\partial L}{\partial \mathbf{C}_i} = \sum_{m=1}^4 \frac{\partial L}{\partial \mathbf{C}_{i,m}} \circ \mathbf{AC}_{i,m} \quad (7)$$

$$\mathbf{C}_i = \mathbf{C}_i - \eta \delta \quad (8)$$

where L denotes the loss function of the network, and η represents the learning rate.

4 Experiments

This section first introduces the datasets, evaluation criteria, and relevant experimental details employed in the experiments. The performance of the proposed method is then compared with those of several existing deep learning-based

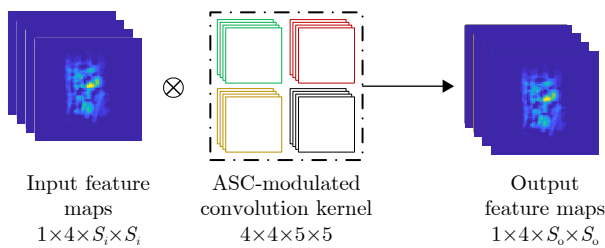


Fig. 4 The convolution process of ASC-modulated convolutional kernel

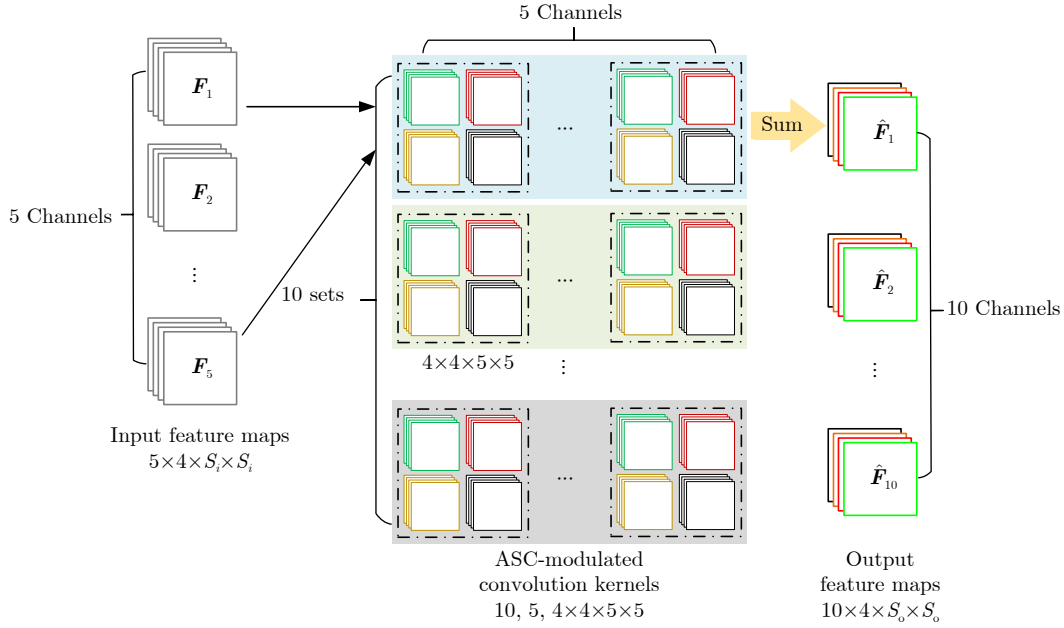


Fig. 5 An example of the complete convolution process in a convolutional layer using ASC-modulated convolution kernels

SAR target recognition approaches to validate the former's effectiveness. Finally, model analyses are conducted, including investigations on the effect of predefined ASC kernel parameters and different training sample numbers on the recognition results. All experimental codes are implemented using Pytorch on an Ubuntu system.

4.1 Introduction to the experimental datasets

The following two datasets are employed in the SAR target recognition experiments: MSTAR^[21] and OpenSARShip^[22] datasets. The MSTAR dataset was acquired by an X-band radar system and contains 10 distinct categories of ground military targets. Each image consists of complex-valued SAR data with a resolution of $0.3 \text{ m} \times 0.3 \text{ m}$ and an image size of $128 \text{ pixel} \times 128 \text{ pixel}$. Fig. 6 displays representative SAR images and their corresponding optical images for all 10 target categories. Compared with optical images, the SAR images in Fig. 6 exhibit substantially poorer visual discriminability.

The SAR images in the MSTAR dataset have an orientation coverage of $0^\circ \sim 360^\circ$. The training set consists of SAR images acquired at a depression angle of 17° , and the test set contains images with a 15° depression angle. Following the established practices in Refs. [23–25], most SAR target recognition methods are validated on two subsets: Three-target and ten-target subsets. The

detailed information on these two subsets is shown in Tabs. 1 and 2. The test sets in both subsets include some variant target types that are not present in the training sets, thus increasing the difficulty of recognition and imposing high demands on the model's feature extraction capability.

The OpenSARShip ship dataset contains 11,346 ship image chips cropped from Sentinel-1 SAR images, as shown in Fig. 7. It features complex backgrounds, including sea clutter, ship wakes, interfering vessels, reefs, and other background disturbances. Moreover, the ship targets exhibit large intraclass variations and relatively small interclass differences, making the OpenSARShip dataset challenging for recognition methods and placing high demands on the model's feature extraction capability.

The OpenSARShip dataset includes two types of image products. In this work, Ground Range Detected (GRD) imagery with a spatial resolution of $10 \text{ m} \times 10 \text{ m}$ is selected for the experiments. Considering the substantial class imbalance in the dataset, we select the classes with a large number of samples—Cargo, Fishing, and Tanker—for the recognition experiments. The detailed information of the three-target dataset used for recognition is shown in Tab. 3.

4.2 Evaluation metrics and experimental details

To verify the effectiveness of the proposed



Fig. 6 Examples of SAR images and corresponding optical images of ten different types of ground targets in MSTAR dataset

Tab. 1 The detailed information of three-target MSTAR data

Category	Type	Number of training samples	Number of testing samples
BMP2	SNC21	233	196
	SN9563	0	195
	SN9566	0	196
BTR70	C71	233	196
	SNS7	0	191
T72	SN132	232	196
	SN812	0	195

Tab. 2 The detailed information of ten-target MSTAR data

Category	Number of training samples	Number of testing samples
BTR60	255	195
2S1	299	274
BRDM2	298	274
D7	299	274
T62	299	273
ZIL131	299	274
ZSU23/4	299	274
BMP2	233	587
BTR70	233	196
T72	232	582

deep network based on ASC-modulated convolutional kernels for SAR target recognition, this work adopts the Probability of Correct Classifica-

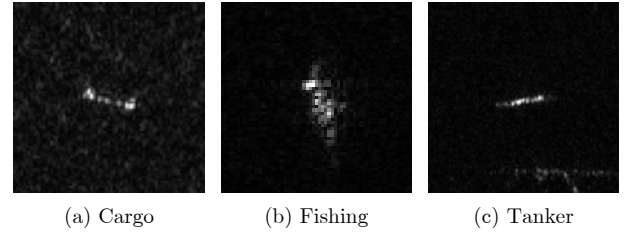


Fig. 7 Examples of SAR images in OpenSARShip dataset

Tab. 3 The detailed information of three-target OpenSARShip data

Category	Number of training samples	Number of testing samples
Cargo	241	159
Fishing	92	23
Tanker	115	76

tion (PCC) to evaluate the target recognition performance of different methods. The calculation formula is as follows:

$$\text{PCC} = \frac{\sum_{i=1}^{N_c} \text{Tr}_i}{\sum_{i=1}^{N_c} Q_i} \quad (9)$$

where Tr_i denotes the number of correctly classified samples in the i -th test class, Q_i represents the total number of samples in the i -th test class, and N_c is the total number of classes. A high PCC indicates good recognition performance.

The proposed network is trained using mini-batch SGD with a batch size of 16. The momentum is set to 0.9, the weight decay factor is 0.0005, and the initial learning rate is 0.005. The learning rate is multiplied by 0.1 every 100 epochs. The total number of training epochs is set to 500 to ensure full network convergence.

4.3 Recognition performance comparison

The performance of the proposed deep network based on ASC-modulated convolutional kernels for SAR target recognition is compared with those of several existing deep learning-based SAR target recognition methods, including A-ConvNet proposed in Ref. [7], VGG16 proposed in Ref. [2], ResNet34 proposed in Ref. [3], and the multiscale CNN based on component analysis (CA-MCNN) proposed in Ref. [26], to verify its effectiveness. In particular, A-ConvNet, VGG16, and ResNet34 are CNN-based SAR target recognition methods. In the original studies, data augmentation was employed to achieve good recognition performance. However, the data augmentation strategies used for SAR images are often directly borrowed from those developed for optical images, which may not be well-suited for SAR data. In practical applications, determining the appropriate augmentation method and the number of augmented samples is difficult. These issues may lead to inappropriate augmentation results, thereby negatively affecting the SAR target recognition performance. By contrast, the proposed network does not suffer from these problems because it does not rely on data augmentation techniques to achieve good recognition performance. Therefore, the recognition results of VGG16, ResNet34, and A-ConvNet are obtained using their original architectures and parameter settings from Refs. [2,3,7], respectively, but without any data augmentation to ensure fairness. These results are then compared with the recognition performance of the proposed method. CA-MCNN treats the ASCs of

SAR targets as components and combines global information from the entire SAR image with the local component information corresponding to ASCs to obtain robust and descriptive features. Moreover, this network utilizes multiscale feature fusion to enhance the representation capability of global features, thereby further improving the final fused features. This study also constructs a baseline network (BaseNet) that shares the same architecture as the proposed network but uses traditional 2D CNN kernels in all convolutional layers. The performance of BaseNet is also compared with that of the proposed method. Different from the proposed method shown in Fig. 1, BaseNet does not require channel expansion. The number of kernels in the first two convolutional layers of BaseNet is set to be equal to the number of ASC-modulated kernel groups multiplied by the number of directional channels to produce the same number of feature maps as the proposed method.

In this section, the recognition performance of the proposed network and the aforementioned methods is compared on three datasets: The three-target MSTAR subset, the ten-target MSTAR subset, and the three-target OpenSAR-Ship dataset.

4.3.1 Recognition experiments on the three-target MSTAR subset

In this section, the effectiveness of our proposed ASC kernel-modulated SAR target recognition network on the three-target MSTAR subset is evaluated. Tab. 4 compares the recognition performance and parameter counts of different methods on this subset.

The results show that our network achieves a PCC of 0.9875 and outperforms VGG16 and Res-

Tab. 4 Recognition performance of different methods on three-target MSTAR data and the number of parameters of each method

Methods	Parameters	PCC
VGG16	6.51×10^7	0.9319
ResNet34	2.13×10^7	0.9253
A-ConvNet	3.03×10^5	0.9385
BaseNet	2.27×10^6	0.9495
CA-MCNN	6.52×10^6	0.9861
Proposed	2.26×10^6	0.9875

Net34 in recognition accuracy. This performance advantage can be attributed to several factors. First, VGG16 and ResNet34 contain substantially more parameters and typically require large-scale training data. In our experiments, only the limited training samples from the three-target subset without data augmentation are used. Second, the test set includes variant types lacking in the training data, thus substantially increasing recognition difficulty. These conditions prevent VGG16 and ResNet34 from fully leveraging their feature extraction capabilities. By contrast, our network's compact architecture requires fewer parameters and demonstrates better performance with limited training data, making it suitable for the three-target MSTAR subset.

Furthermore, the proposed method achieves a higher PCC than A-ConvNet. A CNN's feature extraction capability is generally correlated with its parameter count (more parameters, stronger feature extraction). Despite reducing training data requirements, A-ConvNet's extremely low parameter count ultimately limits its feature extraction capacity, resulting in its suboptimal performance on the three-target MSTAR subset. By contrast, our network employs ASC-modulated convolution kernels that better capture SAR-specific scattering structures and edge features, leading to superior recognition performance. Compared with BaseNet, our network requires fewer parameters and delivers a higher PCC. This advantage stems from our ASC modulation mechanism: By learning only a small set of base CNN kernels and generating additional kernels through ASC modulation, the proposed method is able to maintain strong feature extraction capability while reducing its parameters. The benefits of ASC modulation persist throughout training iterations without diminishing. Tab. 4 shows that CA-MCNN outperforms other comparative methods by leveraging image data and ASC information. However, our approach achieves higher PCC with fewer parameters without requiring per-image ASC extraction, demonstrating advantages in simplicity and performance.

A comparative analysis of parameter counts

and performance is performed between the proposed method and other approaches in Tab. 4. Compared with VGG16 and ResNet34, the proposed method substantially reduces the number of parameters and improves recognition performance. Compared with A-ConvNet and BaseNet, the proposed method achieves better recognition performance with comparable parameter counts. While Tab. 4 only shows its parameter count, CA-MCNN additionally requires ASC extraction for each SAR image. By contrast, the proposed method can greatly reduce the parameter count while maintaining comparable performance. In summary, by incorporating the electromagnetic scattering characteristics of SAR targets, the proposed approach can either have a substantially improved recognition performance with a similar parameter count or a substantially reduced parameter count with a similar recognition performance to the other methods.

The experimental results on the three-target MSTAR subset demonstrate the effectiveness of our proposed network. By employing ASC-modulated convolution kernels to extract scattering structures and edge features aligned with SAR target characteristics, the network maintains strong feature extraction capability while reducing parameter count, thereby further decreasing its dependence on the number of training samples.

4.3.2 Recognition experiments on the ten-target MSTAR subset

This section further discusses the effectiveness of the proposed ASC kernel-modulated deep network for SAR target recognition on the ten-target MSTAR subset. Tab. 5 presents the recognition performance and parameter counts of different methods on this subset.

First, Tabs. 4 and 5 reveal the degraded performance of all the methods on the ten-target subset relative to that on the three-target subset. This phenomenon occurs because the ten-target subset contains variant types absent from the training data and seven additional military target categories, each with limited training samples. Therefore, the ten-target subset presents great recognition challenges, leading to reduced perform-

Tab. 5 Recognition performance of different methods on ten-target MSTAR data and the number of parameters of each method

Methods	Parameters	PCC
VGG16	6.51×10^7	0.9166
ResNet34	2.13×10^7	0.9138
A-ConvNet	3.03×10^5	0.9219
BaseNet	2.27×10^6	0.9422
CA-MCNN	6.52×10^6	0.9781
Proposed	2.26×10^6	0.9844

ance for all the methods. Second, Tab. 5 shows that our method achieves a PCC of 0.9844. Consistent with the findings in Tab. 4, the proposed method outperforms the parameter-heavy approaches (VGG16 and ResNet34) and parameter-light methods (A-ConvNet) while surpassing BaseNet and CA-MCNN in PCC. This advantage also stems from our ASC-modulated kernels, which can generate multiple convolution kernels from a small, learned set through ASC modulation and capture SAR-specific scattering structures and edge features. These properties enable our network to maintain strong feature extraction capability with reduced parameters, achieving superior performance on the data-limited ten-target MSTAR subset.

The detailed comparison of parameter counts and performance metrics between the proposed method and other approaches in Tab. 5 yields conclusions consistent with those from Tab. 4. By effectively incorporating the electromagnetic scattering characteristics of SAR targets, our method demonstrates substantial advantages over comparative approaches: it either achieves a substantially improved recognition performance with a comparable parameter count or maintains similar recognition accuracy with a dramatically reduced parameter count.

The experimental results on the ten-target MSTAR subset further validate the effectiveness of our proposed network. By utilizing ASC-modulated convolution kernels to extract scattering structures and edge features matching SAR target characteristics, the network maintains robust feature extraction capability while reducing parameter count. It also demonstrates excellent recognition performance even when handling challenging datasets of great target variety.

4.3.3 Recognition experiments on the OpenSARShip ship dataset

In this section, the effectiveness of the proposed ASC kernel-modulated deep network for SAR target recognition on the OpenSARShip dataset is examined. Given that CA-MCNN requires ASC extraction using complex-valued data with true radar resolution and bandwidth parameters—conditions that neither of OpenSARShip's two image products can satisfy—it is excluded from the comparative analysis in this section. Tab. 6 presents the recognition performance and parameter counts of different methods on the three-target OpenSARShip dataset.

First, a comparative analysis with the MSTAR dataset results reveals the degraded recognition performance of all the methods for the three-target OpenSARShip dataset. This performance reduction stems from OpenSARShip's complex background conditions and the inherent characteristics of ship targets—large intraclass variations and relatively small interclass differences—which collectively increase recognition difficulty. As shown in Tab. 6, our proposed network achieves a superior PCC of 0.8101 on the three-target OpenSARShip dataset compared with the alternative methods. The performance limitation manifests differently across the architectures: (1) VGG16 and ResNet34 underperform due to the insufficient training data for their extensive parameters; (2) A-ConvNet's constrained parameter count limits its feature extraction capacity for these challenging ship targets; and (3) compared with that of BaseNet, our proposed method's higher PCC confirms that ASC-modulated kernels extract more SAR-appropriate features than conventional convolutional layers. As a consequence, the proposed method still achieves

Tab. 6 Recognition performance of different methods on three-target OpenSARShip data and the number of parameters of each method

Methods	Parameters	PCC
VGG16	6.51×10^7	0.7713
ResNet34	2.13×10^7	0.7403
A-ConvNet	3.03×10^5	0.7791
BaseNet	2.27×10^6	0.8062
CA-MCNN	6.52×10^6	—
Proposed	2.26×10^6	0.8101

optimal recognition performance on the complex OpenSARShip dataset, demonstrating its robustness for SAR target recognition in challenging background conditions.

4.4 Analysis of the proposed model

4.4.1 Impact of the number of predefined ASC kernel directions on recognition performance

Section 3.2 presents the analysis of the selection of orientation angles $\bar{\varphi}$ for the distributed ASCs. To extract scattering structures and edge features from SAR images in different directions, the proposed method uniformly samples orientation angles $\bar{\varphi}$ in the range of $[-90^\circ, 90^\circ]$. Assuming that a total of M orientation angles are sampled, M -distributed ASCs with different orientation angles can be obtained. Although a large number of orientation angles $\bar{\varphi}$ provides diverse distributed ASC representations, it also increases the number of network parameters. The proposed method balances parameter complexity and directional richness by setting $M = 4$, *i.e.*, $\bar{\varphi} = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$. To further analyze the impact of the number of predefined ASC kernel orientation angles on recognition performance, we evaluate the proposed method's recognition performance on the three-target MSTAR subset under different M values. The results are shown in Tab. 7.

As shown in Tab. 7, the number of predefined ASC kernel orientations M has a substantial impact on recognition performance. When the number of orientations is too small, such as $M = 2$, the predefined ASC kernels lack sufficient diversity in orientation. Extracting scattering structures and edge features from different directions becomes difficult, which in turn leads to a degraded recognition performance. When the number of orientations is too large, such as $M = 6$, the diversity of the predefined ASC ker-

nels is ensured. However, the increased number of network parameters may negatively affect the recognition performance when the training samples are limited. Tab. 7 shows that when $M = 4$, the proposed network achieves the highest recognition performance on the three-target MSTAR subset. When $M = 4$, $\bar{\varphi} = [-45^\circ, 0^\circ, 45^\circ, 90^\circ]$, the predefined ASC kernels strike a balance between orientation diversity and parameter efficiency. In summary, Tab. 7 demonstrates the rationale of setting the number of predefined ASC kernel orientations to $M = 4$ in the proposed method.

4.4.2 Impact of the predefined ASC kernel length on recognition performance

Section 3.2 presents the analysis of the selection of the length of the distributed ASCs. The length is jointly determined by the spatial resolution of the image and the convolution kernel size of the CNN. Given that the spatial resolution of the MSTAR dataset is $0.3 \text{ m} \times 0.3 \text{ m}$, the length can be set according to the convolution kernel size. For example, with a 5×5 convolution kernel, the length L can be set to $[0.3, 0.6, 0.9, 1.2]$. In accordance with the proposed method's aim of extracting scattering structures and edge features that reflect the characteristics of SAR images through the ASC modulation kernel, the length of the distributed ASCs in the predefined ASC kernel should not be too short. To ensure the completeness of the distributed ASCs within the ASC kernel and based on the visualization results of ASC kernels shown in Fig. 2, we select the ASC kernel corresponding to $L = 0.9$ as the predefined ASC kernel. In this section, the recognition performance of the proposed method on the three-target MSTAR subset under different lengths is evaluated to further analyze the impact of the length of predefined distributed ASCs on recognition performance. The results are reported in Tab. 8.

As shown in Tab. 8, the length of the predefined ASC kernel has a substantial impact on recognition performance. When the length is too short, such as 0.3, the ASC kernel fails to effectively extract the scattering structures and edge

Tab. 7 Recognition performance on three-target MSTAR data under different values of M

M	PCC
2	0.9722
3	0.9817
4	0.9875
5	0.9810
6	0.9744

features in SAR images, leading to degraded recognition performance. When the length is set to 1.2 (ASC kernel visualization in Fig. 2), the completeness of the distributed ASCs within the ASC kernel cannot be guaranteed, which also negatively affects recognition performance. When the length is set to 0.9, the proposed network achieves the highest recognition performance on the three-target MSTAR subset. At this length, the network can effectively extract scattering structures and edge features that reflect SAR image characteristics while maintaining the integrity of the distributed ASCs within the ASC kernel. In summary, Tab. 8 presents the rationale for setting the predefined ASC kernel's length to 0.9 in the proposed method.

4.4.3 Impact of using ASC-modulated convolutional kernels in different layers on recognition performance

The proposed network consists of five convolutional layers and two fully connected layers. The shallow layers of CNNs are responsible for extracting scattering structures and edge features from images. Hence, the proposed method applies the ASC-modulated convolution kernels in the first two convolutional layers to extract the structures and features reflecting the characteristics of SAR images. The remaining three convolutional layers use standard CNN kernels to ensure the extraction of discriminative semantic features. The proposed ASC-modulated convolution kernel is a plug-and-play module that can be directly used to replace convolutional layers in deep networks such as VGG and ResNet. The convolutional layers in VGG16, ResNet18, and the proposed method are selectively replaced with ASC-modulated convolution kernels. Their recognition performance on the three-target MSTAR subset is then evaluated to further analyze the impact of ASC-modulated convolution kernels applied in differ-

ent layers of various network architectures on their recognition performance. The results are listed in Tab. 9.

First, Tab. 9 shows that the applied ASC-modulated convolution kernels improve the recognition performance of all three networks, indicating that the ASC-modulated convolution kernels can serve as a plug-and-play module to enhance SAR target recognition performance in deep networks. Second, applying the ASC-modulated convolution kernels in the two shallowest convolutional layers yields a good recognition performance. Introducing ASC-modulated convolution kernels in shallow layers can inject the electromagnetic characteristics of SAR images into early-stage features, making them suitable for SAR target recognition. Therefore, the ASC-modulated convolution layers should be applied starting from the first convolutional layer of the network. Finally, the number of convolutional layers applied with ASC-modulated convolution kernels also affects recognition performance. When only the first convolutional layer uses ASC-modulated kernels, the extraction of scattering structures and edge features from SAR images is insufficient. When all four convolutional layers adopt ASC-modulated kernels, the network's ability to extract semantic features may be compromised. Both cases negatively affect recognition performance. As shown in Tab. 9, the proposed network achieves the highest recognition performance on the three-target MSTAR subset when ASC-modulated convolution kernels are applied on the first two convolutional layers.

In summary, when using ASC-modulated convolution kernels as a plug-and-play module, the

Tab. 8 Recognition performance on three-target MSTAR data under different values of L

L	PCC
0.3	0.9729
0.6	0.9788
0.9	0.9875
1.2	0.9810

Tab. 9 Recognition performance on three-target MSTAR data under different convolution layers using ASC modulated convolutional kernels

Convolution layers used	VGG16	ResNet18	Proposed
Without ASC	0.9319	0.948	0.9495
First layer	0.9795	0.9663	0.9795
First & Second layers	0.9839	0.9769	0.9875
First to Third layers	0.9773	0.9736	0.9810
First to Fourth layers	0.9758	0.9729	0.9736
Second & Third layers	0.9766	0.9641	0.9751
Third & Fourth layers	0.9714	0.9582	0.9707

application should start from the first layer. The specific number of layers to which these kernels are applied should be based on the network architecture and the available training data. The experimental results presented in this paper can serve as a reference for making such selections.

4.4.4 Impact of different numbers of training samples on recognition performance

The proposed network employs ASC-modulated convolution kernels, which can extract scattering structures and edge features reflecting the characteristics of SAR images. An ASC-modulated convolution kernel is obtained by modulating a set of learnable CNN kernels with a pre-defined set of ASC kernels. This process allows the generation of a large number of convolution kernels from only a small set of learnable CNN kernels, thereby maintaining the CNN's feature extraction capability while reducing the number of parameters. As a consequence, the demand for training samples is minimized. To verify that the proposed network can maintain high recognition performance while reducing the dependence on training data, we examine its recognition performance on the three-target MSTAR subset under varying numbers of training samples. The results are shown in Tab. 10.

Tab. 10 show the performance of different SAR target recognition methods under varying training sample sizes of 100%, 50%, 30%, 25%, and 20%. First, VGG16 and ResNet34 exhibit strong reliance on training data due to their large number of parameters. As a consequence, their recognition performance drops substantially as the number of training samples decreases. Second, A-ConvNet and BaseNet with fewer parameters outperform VGG16 and ResNet34 when the training samples are limited. However, A-ConvNet and

BaseNet rely solely on SAR image information, so their recognition performance still depends heavily on the completeness of training data. Thus, their performance also degrades as the number of training samples decreases. In contrast to the four methods mentioned above, CA-MCNN maintains strong recognition performance even when few training samples are available. This method leverages SAR image information and ASC information to learn many descriptive features, thereby improving recognition performance under limited data conditions. Compared with CA-MCNN, the proposed network achieves better performance when the number of training samples is further reduced. When only 25% of the training samples are used, the proposed method still achieves a PCC of 0.9187. This phenomenon is attributed to the use of ASC-modulated convolution kernels, which reduce the number of network parameters and extract scattering structures and edge features reflecting SAR image characteristics. As a consequence, the proposed method maintains excellent recognition performance even in data-scarce scenarios. In summary, Tab. 10 demonstrates the effectiveness of the proposed deep network based on ASC kernel modulation for SAR target recognition. The kernel modulation reduces reliance on training samples while ensuring high recognition performance.

5 Conclusion

This work investigates a deep network based on ASC kernel modulation for SAR target recognition. By integrating the electromagnetic scattering characteristics of SAR targets, a simple network structure is designed to extract features aligned with the properties of SAR images. This approach reduces the number of network para-

Tab. 10 Recognition performance of different methods on three-target MSTAR data with different number of training samples

Methods	Training sample ratio				
	100%	50%	30%	25%	20%
VGG16	0.9319	0.8886	0.8352	0.7883	0.7370
ResNet34	0.9253	0.8813	0.8278	0.7590	0.7267
A-ConvNet	0.9385	0.9062	0.8769	0.8337	0.7875
BaseNet	0.9495	0.8974	0.8674	0.8227	0.7758
CA-MCNN	0.9861	0.9641	0.9165	0.8938	0.8608
Proposed	0.9875	0.9670	0.9480	0.9187	0.8711

meters while preserving the feature extraction capability, thereby maintaining excellent recognition performance under limited training data. First, the proposed method introduces ASC-modulated convolution kernels, which modulate the original CNN kernels using a predefined set of ASC kernels with different orientations and lengths. This process enables the generation of a large set of convolution kernels from a small number of learnable ones. Second, the ASC-modulated convolution kernels are applied only in the shallow network layers to extract scattering structures and edge features that are consistent with SAR image characteristics. The deeper layers still use standard CNN kernels to extract semantic features. By combining ASC-modulated kernels with standard CNN kernels throughout the network, the proposed architecture captures the electromagnetic scattering properties of SAR targets and the feature extraction advantages of results on the MSTAR and OpenSARShip ship datasets demonstrate that the proposed network simultaneously achieves high recognition performance and reduced dependence on training samples.

Conflict of Interests The authors declare that there is no conflict of interests

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